Indian Institute of Information Technology, Surat
भारतीय सूचना प्रौद्योगिकी संस्थान,सूरत
(Institute of National Importance under Act of Parliament)

EC 503: Image Processing and Computer Vision

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Gradient operator

- First derivatives are implemented using the magnitude of the gradient

$$\nabla \mathrm{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

$$\nabla f = mag(\nabla \mathrm{f}) = [G_x^2 + G_y^2]^{1/2}$$

$$= \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

commonly approx

$$\nabla f \approx |G_x| + |G_y|$$

the magnitude becomes nonlinear

Gradient mask

- Simplest approximation, 2 x 2

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$G_x = (z_8 - z_5) \quad \text{and} \quad G_y = (z_6 - z_5)$$

$$\nabla f = [G_x^2 + G_y^2]^{1/2} = [(z_8 - z_5)^2 + (z_6 - z_5)^2]^{1/2}$$

$$\nabla f \approx |z_8 - z_5| + |z_6 - z_5|$$

Gradient mask

- Roberts cross-gradient operators, 2 x 2

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$G_x = (z_9 - z_5) \quad \text{and} \quad G_y = (z_8 - z_6)$$

$$\nabla f = [G_x^2 + G_y^2]^{1/2} = [(z_9 - z_5)^2 + (z_8 - z_6)^2]^{1/2}$$

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$$

−1	0
0	1

0	−1
1	0

Gradient mask

- Sobel operators, 3 x 3

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

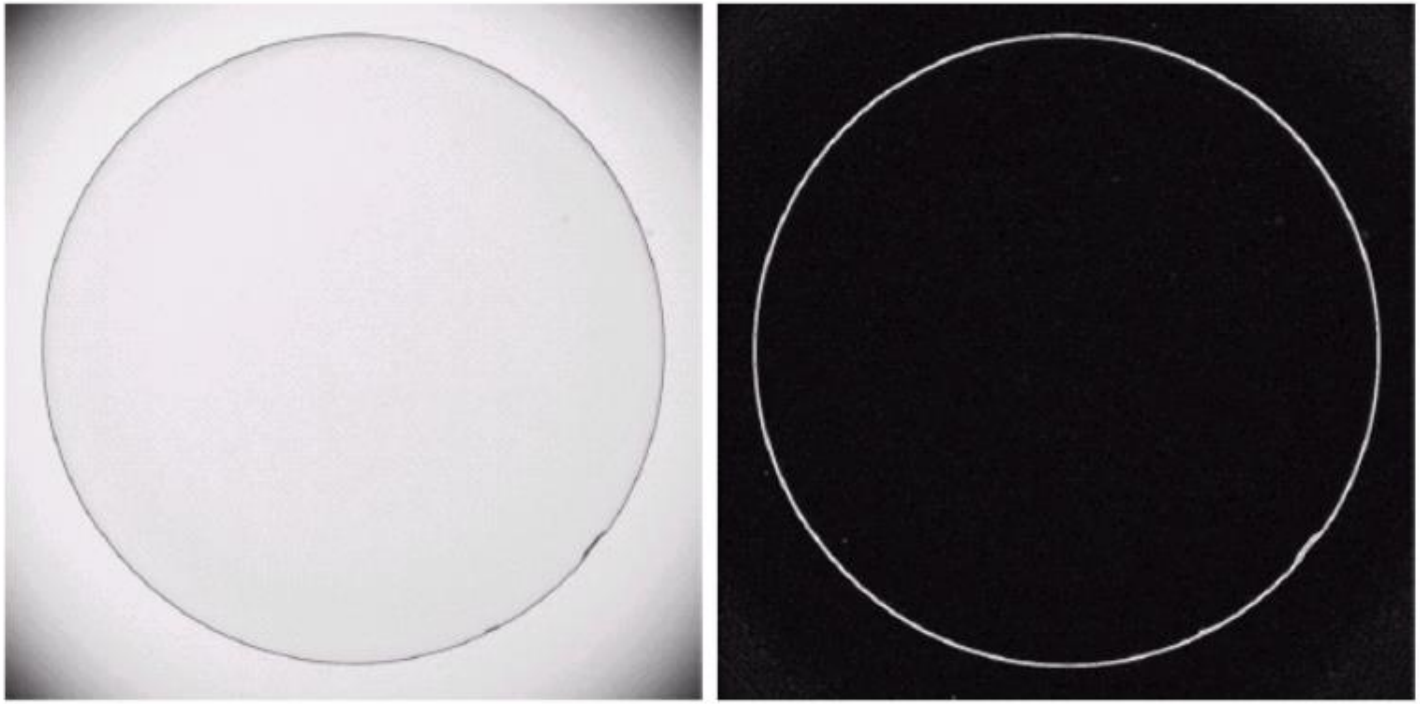
$$\nabla f \approx |G_x| + |G_y|$$

the weight value 2 is to achieve smoothing by giving more important to the center point

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

Example

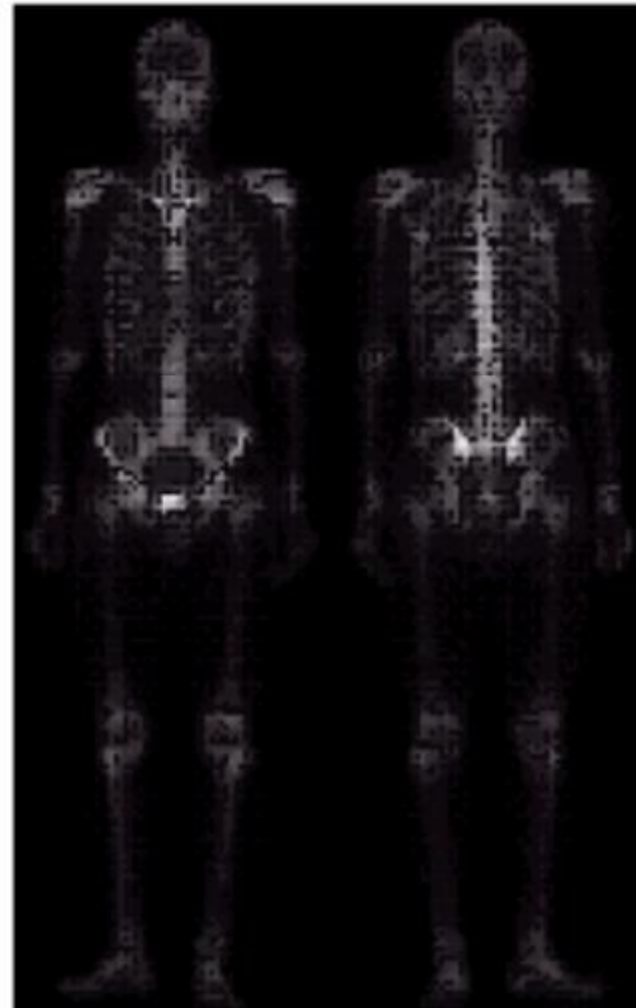


a b

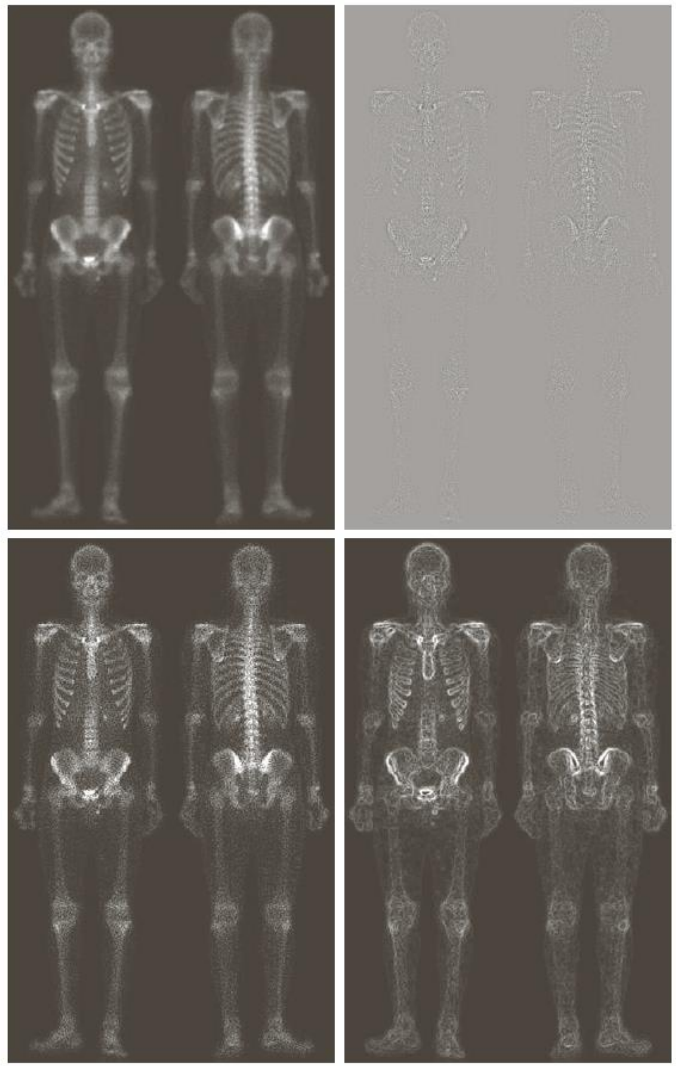
FIGURE 3.45 Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Mr. Pete Sites, Perceptics Corporation.)

Example of Combining Spatial Enhancement Methods

- Want to sharpen the original image and bring out more skeletal detail
- **Problems**:
- Narrow dynamic range of gray level and high noise level and high noise content makes the image difficult to enhance

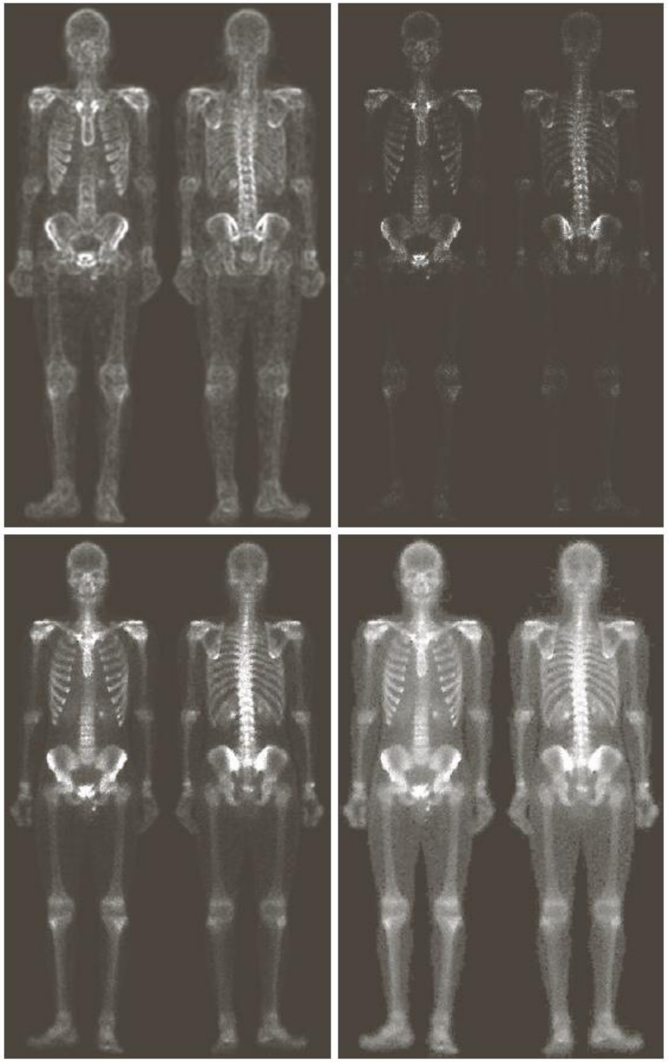


Combining Spatial Enhancement Methods



a b
c d

FIGURE 3.43
(a) Image of whole body bone scan. (b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of (a).



e	f
g	h

FIGURE 3.43 *(Continued)* (e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e). (g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

UNIT2_2:

Image Enhancement in the Frequency Domain

Fourier Transform

- **Fourier Series:** Any function that periodically repeats itself can be expressed as the sum of sines/cosines of different frequencies, each multiplied with a different coefficient

- **Fourier Transform:** Functions that are not periodic, whose area under the curve is finite, can be expressed as the integral of sines and/cosines multiplied by a weighting function

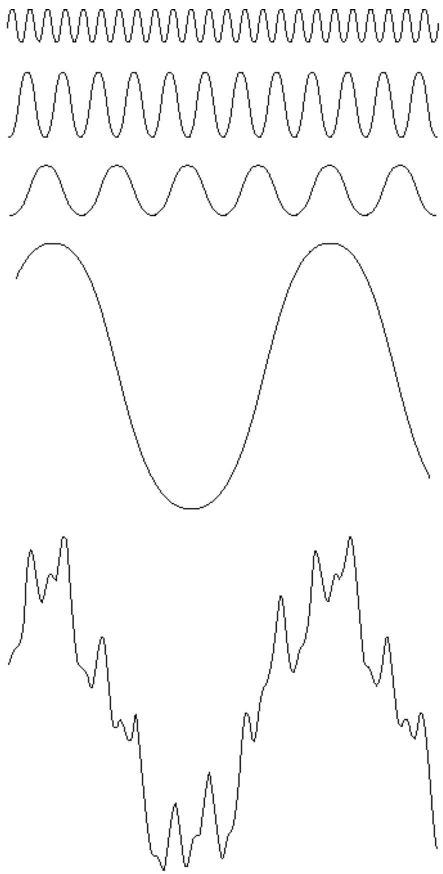


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

1D Continuous Fourier Transform

•*The **Fourier Transform** is an important tool in Image Processing, and is directly related to **filter theory**, since a filter, which is a convolution in the spatial domain, is a simple multiplication in the frequency domain.*

•***1-D Continuous Fourier Transform***

The Fourier transform, F(u), of a single variable continuous function, f(x), is defined by:

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

Given Fourier transform of a function F(u), the inverse Fourier transform can be used to obtain, f(x), by:

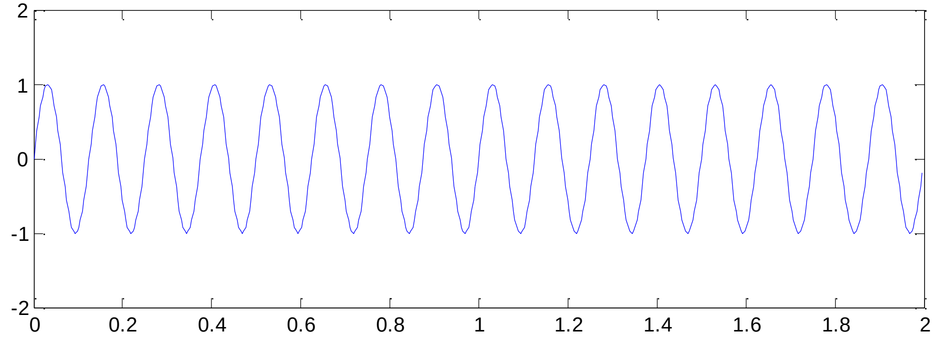
$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$

Note: F(u), is the frequency spectrum, where, u represents the frequency, and f(x) is the

signal where x represents time/space. $j = \sqrt{-1}$

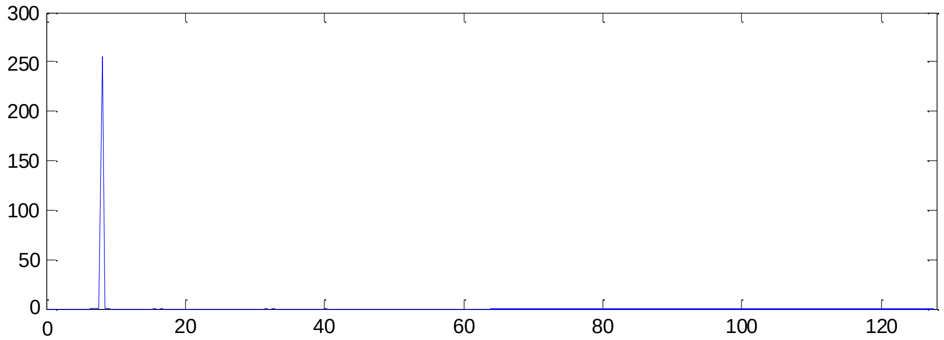
1D Continuous Fourier Transform

Time/Space domain



Sine wave

Frequency domain

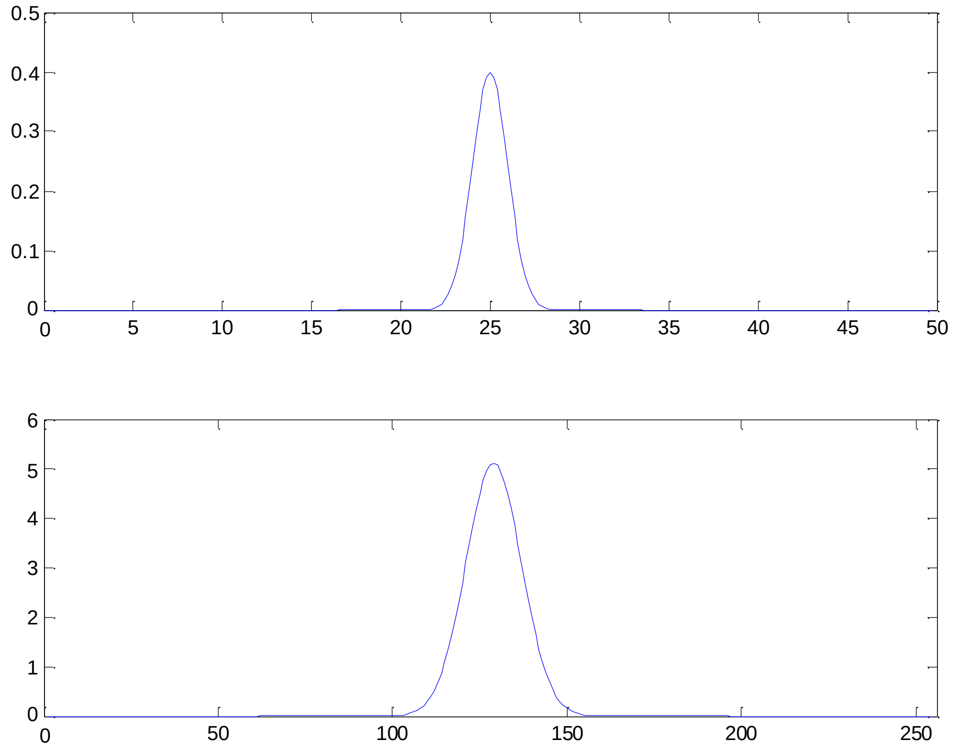


Delta function

Continuous Fourier Transform

Time/Space domain

Gaussian

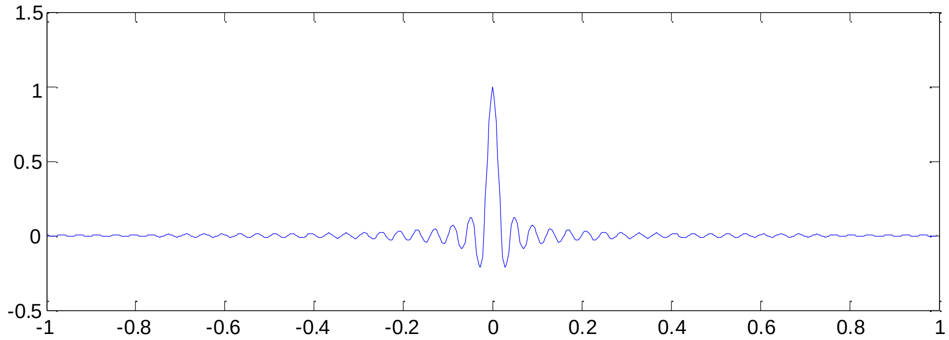


Frequency domain

Gaussian

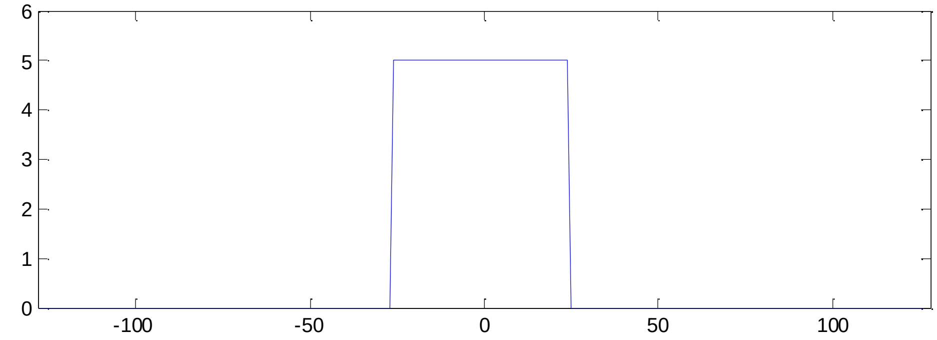
1D Continuous Fourier Transform

Time/Space domain



Sinc function

Frequency domain



Square wave

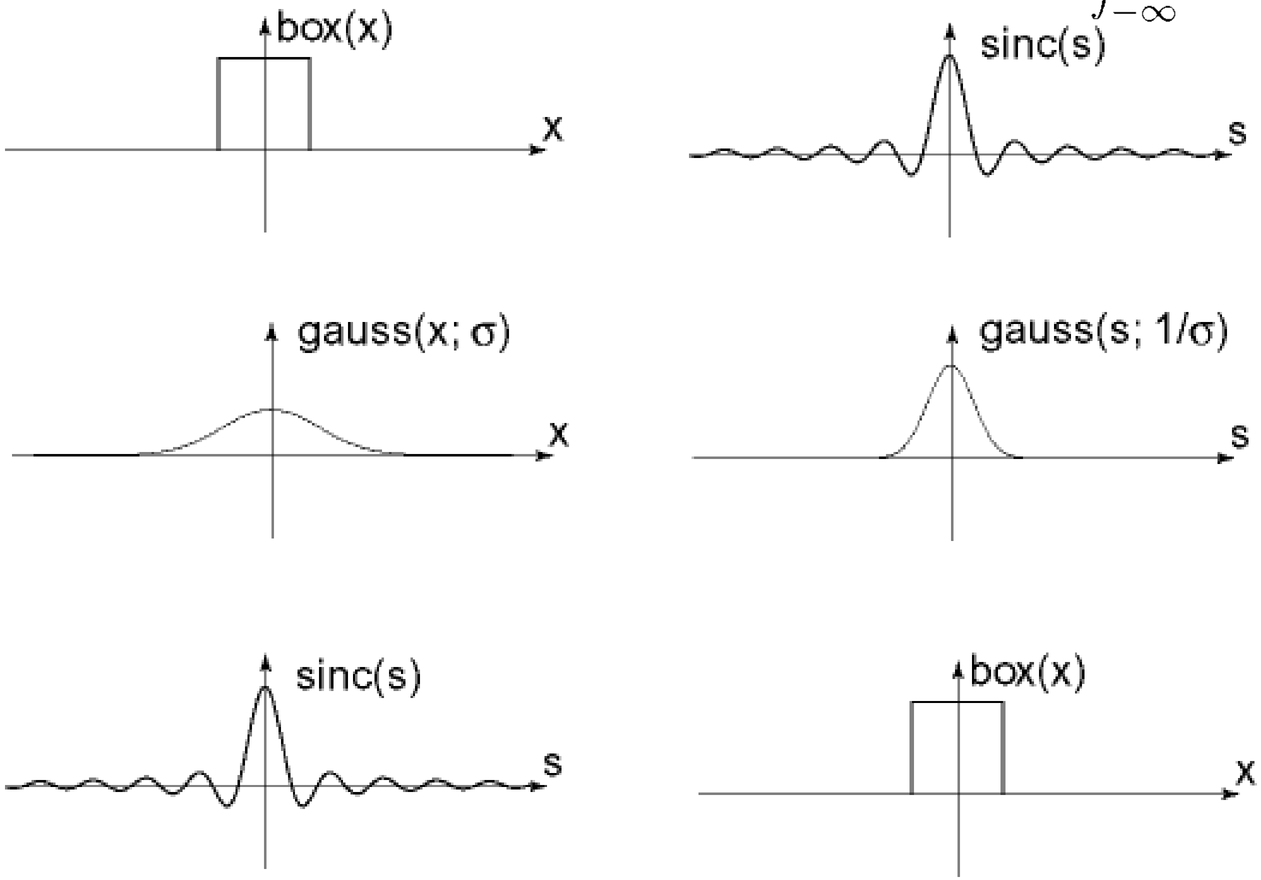
Fourier Transform pairs (spatial versus Frequency)

Spatial domain

$f(x)$

Frequency domain

$$F(s) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi sx} dx$$



1D Discrete Fourier Transform

The Fourier transform, F(u), of a discrete function of one variable, f(x), ***x=0, 1, 2, …, M-1, is given by:***

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M}$$

The inverse Discrete Fourier Transform can be used to calculate f(x), by:

$$f(x) = \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M}$$

Note: F(u), which is the Fourier transform of f(x) contains discrete complex quantities and it has the same number of components as f(x).

$$e^{j\theta} = \cos\theta + j\sin\theta$$

1D Discrete Fourier Transform

Then;

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)[\cos 2\pi ux/M - j\sin 2\pi ux/M]$$

•*The Fourier Transform generates complex quantities. The magnitude or the spectrum of the Fourier transform is given by:*

$$|F(u)| = \left[R^2(u) + I^2(u)\right]^{1/2}$$

R(u) is the Real Part and

I(u) is the Imaginary Part

•*The Phase Spectrum of the transform refers to the angles between the real and imaginary components and it is denoted by:*

$$\phi(u) = \tan^{-1}\left[\frac{I(u)}{R(u)}\right]$$

1D Discrete Fourier Transform

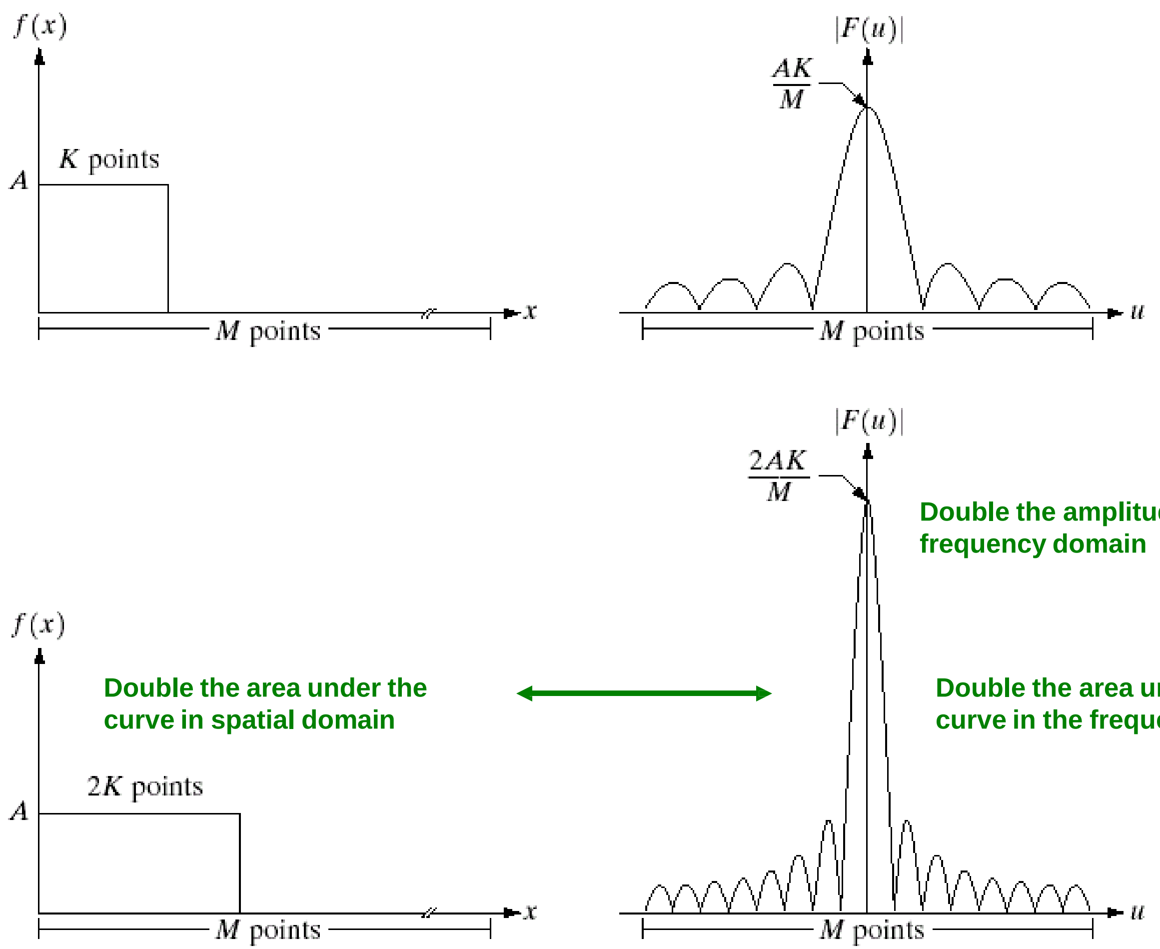
- The **Power Spectrum/spectral density** is defined as the square of the Fourier spectrum and denoted by

$$P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$

- The power spectrum can be used, for example to separate a portion of a specified frequency (i.e. low frequency) power from the power spectrum and monitor the effect. Typically used to define the cut off frequencies used in lowpass and highpass filtering

- We primarily use the Fourier Spectrum for image enhancement applications

1D Discrete Fourier Transform



a b
c d

FIGURE 4.2 (a) A discrete function of M points, and (b) its Fourier spectrum. (c) A discrete function with twice the number of nonzero points, and (d) its Fourier spectrum.

2D Discrete Fourier Transform

The Fourier transform of a 2D discrete function (image) f(x,y) of size MxN is given by:

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

u=0, 1, 2, …, M-1, and v=0, 1, 2, …, N-1 and the inverse 2D Discrete Fourier Transform can be calculated by:

$$f(x,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(ux/M + vy/N)}$$

x=0, 1, 2, …, M-1, and y=0, 1, 2, …, N-1.

2D Discrete Fourier Transform

The 2D Fourier Spectrum, Phase Spectrum and Power Spectrum can be respectively denoted by:

$$|F(u,v)| = \left[R^2(u,v) + I^2(u,v)\right]^{1/2}$$

Magnitude/Fourier Spectrum

$$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$$

Phase Spectrum

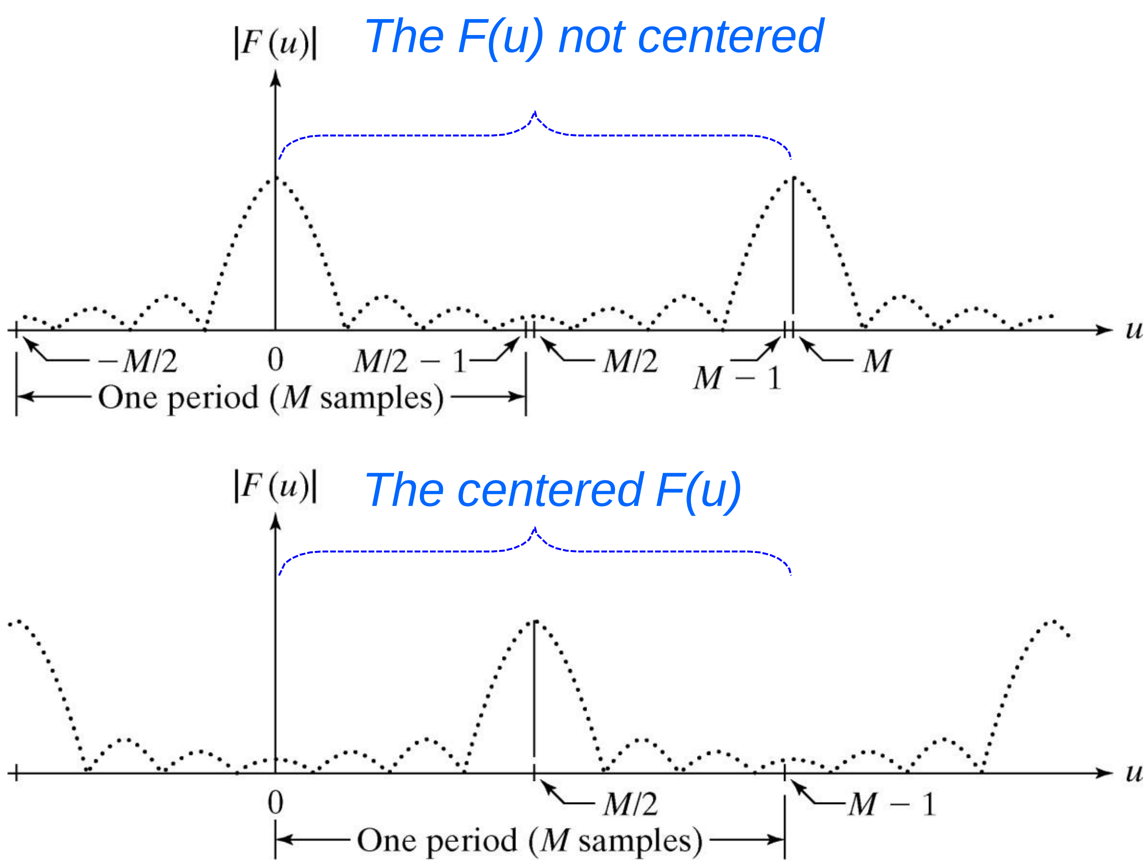
$$P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v)$$

Power Spectrum

2D Discrete Fourier Transform

The Periodicity property: F(u) in 1D DFT has a period of M

The Symmetry property: The magnitude of the transform is centered on the origin.



a
b

FIGURE 4.1 (a) Fourier spectrum showing back-to-back half periods in the interval $[0, M-1]$. (b) Centered spectrum in the same interval, obtained by multiplying $f(x)$ by $(-1)^x$ prior to computing the Fourier transform.

2D Discrete Fourier Transform

The Periodicity property: F(u,v) in 2D DFT has a period of N in horizontal and M in vertical directions

The Symmetry property: The magnitude of the transform is centered on the origin.

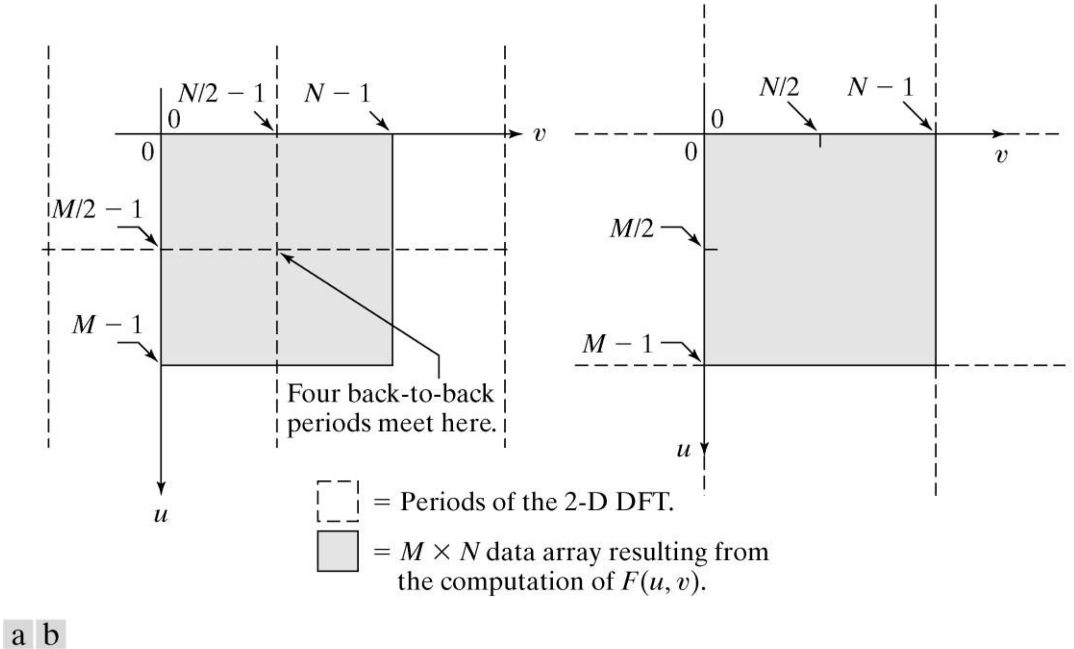
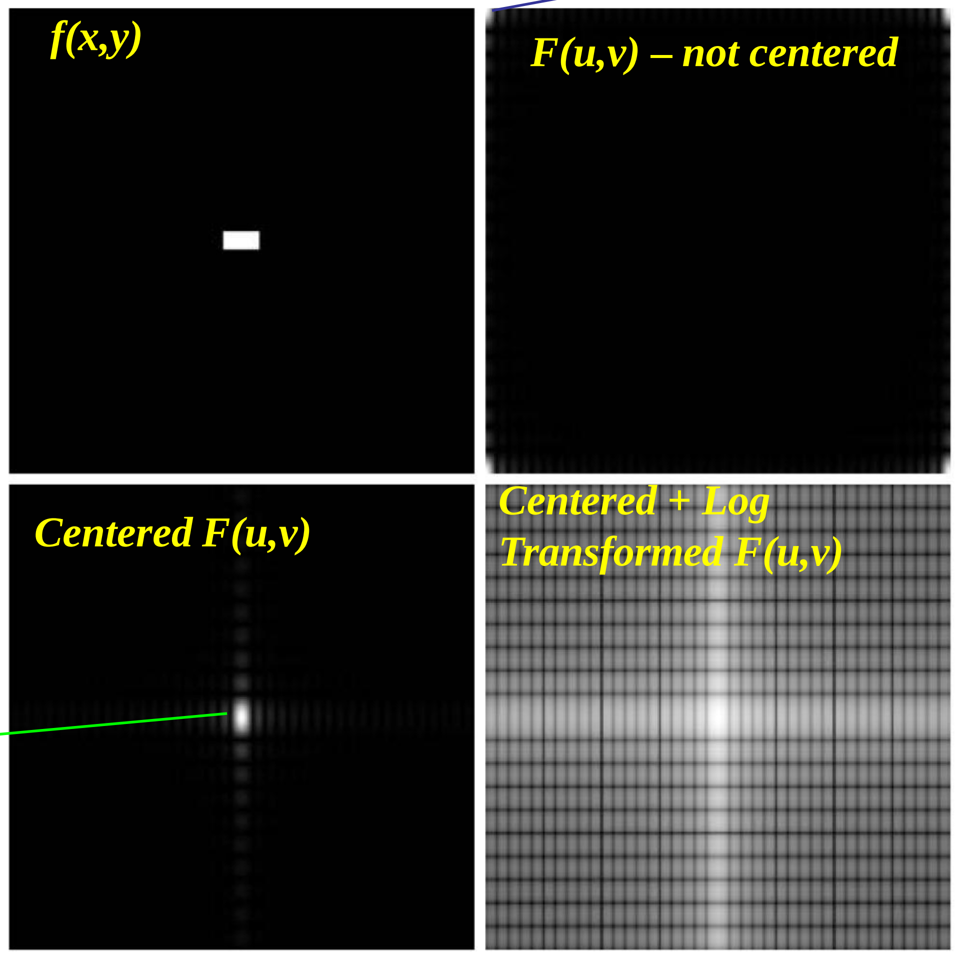


FIGURE 4.2 (a) $M \times N$ Fourier spectrum (shaded), showing four back-to-back quarter periods contained in the spectrum data. (b) Spectrum obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ prior to computing the Fourier transform. Only one period is shown shaded because this is the data that would be obtained by an implementation of the equation for $F(u, v)$.

2D Discrete Fourier Transform

Origin = 0,0



Origin = 0,0

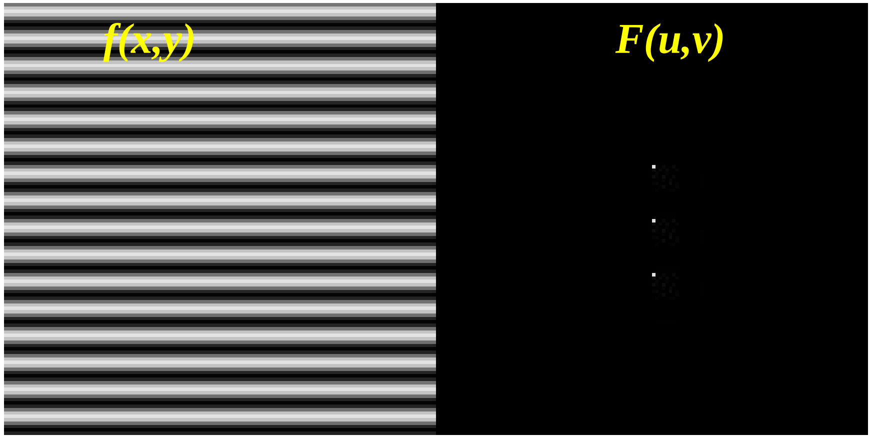
a	b
c	d

FIGURE 4.3
(a) A simple image.
(b) Fourier spectrum.
(c) Centered spectrum.
(d) Spectrum visually enhanced by a log transformation.

2D Discrete Fourier Transform

Consider the following 2 left images which are pure vertical cosine of 4 cycles and a pure vertical cosine of 32 cycles.

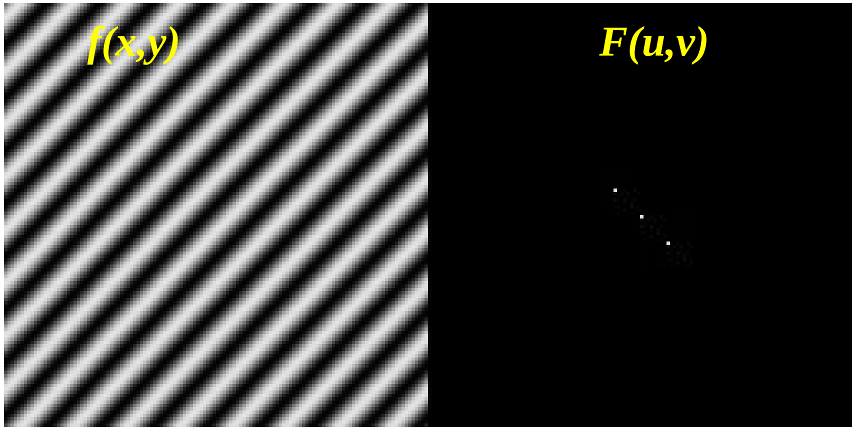
Notice that the Fourier Spectrum for each image at the right contains just a single component, represented by 2 bright spots symmetrically placed about the center.



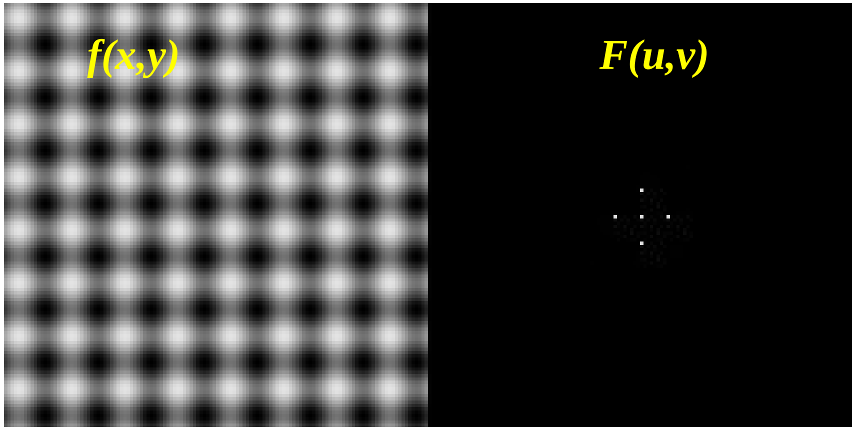
The dot at the center that represents the (0,0) frequency term or average value of the image. Images usually have a large average value/DC component. Due to low frequencies Fourier Spectrum images usually have a bright blob of components near the center.

2D Discrete Fourier Transform

Consider the following 2 left images with pure cosines in pure forward diagonal and mixed vertical+horizontal.



One of the properties of the 2D FT is that if you rotate the image, the spectrum will rotate in the same direction



The sum of 2 sine functions, each in opposite (vertical and horizontal) direction.

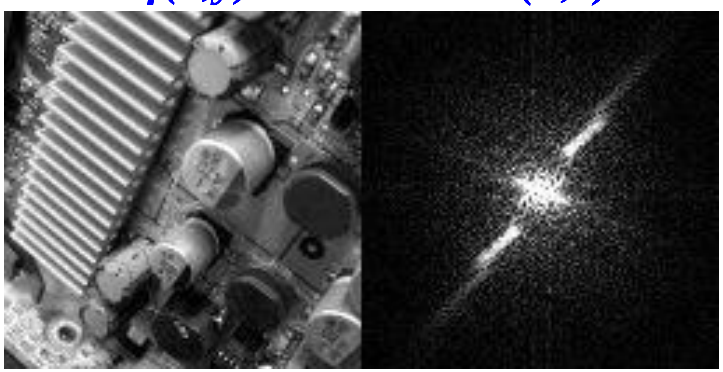
2D Discrete Fourier Transform

*The center value (at the origin) of the Frequency Spectrum corresponds to the **ZERO frequency** component which also referred to as the **DC component** in an image:*

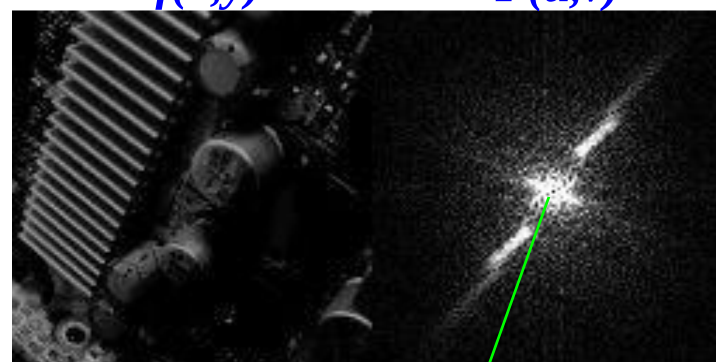
Substituting 0,0 to the origin, the Fourier transform function yields to the average/DC component value as follows:

$$F(0,0) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)$$

f(x,y) ***F(u,v)***



f(x,y) ***F(u,v)***

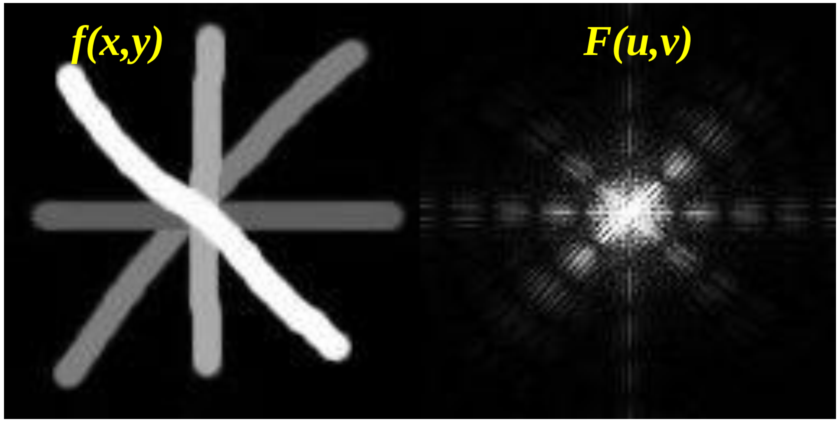


The center/zero frequency component (DC component) is removed

f(x,y)
(Average image)

F(u,v)
(DC Component)

2D Discrete Fourier Transform



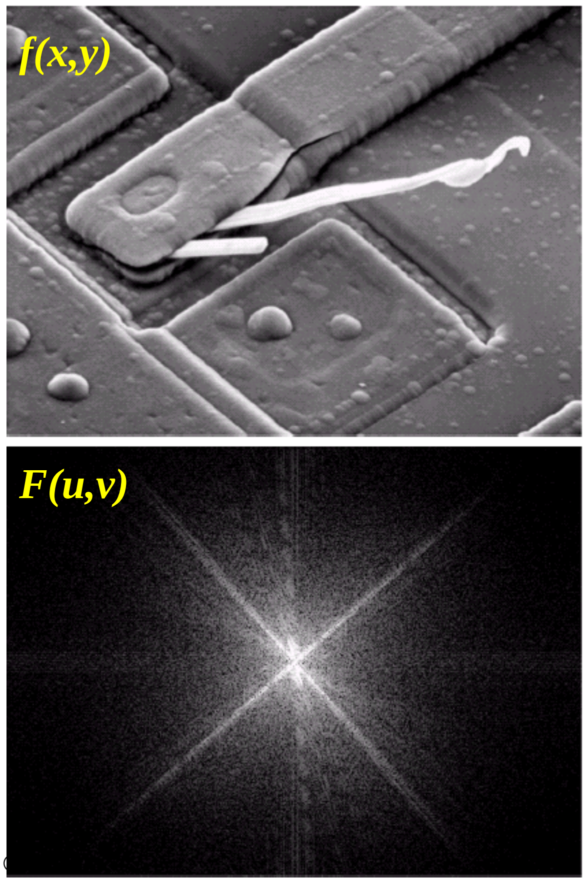
The lines in an image often generate perpendicular lines in the spectrum

The sloped lines in the spectrum are due to the sharp transition from the sky to the mountain

Filtering in the Frequency Domain

•*Some Basic Properties of the Frequency Domain:*

•*Frequency is directly related to the rate of change. Therefore, slowest varying component (u=v=0) corresponds to the average intensity level of the image. Corresponds to the origin of the Fourier Spectrum.*



a
b

FIGURE 4.4 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

•*Higher frequencies corresponds to the faster varying intensity level changes in the image. The edges of objects or the other components characterized by the abrupt changes in the intensity level corresponds to higher frequencies.*

Filtering in the Frequency Domain

•*Basic Steps for Filtering in the Frequency Domain:*

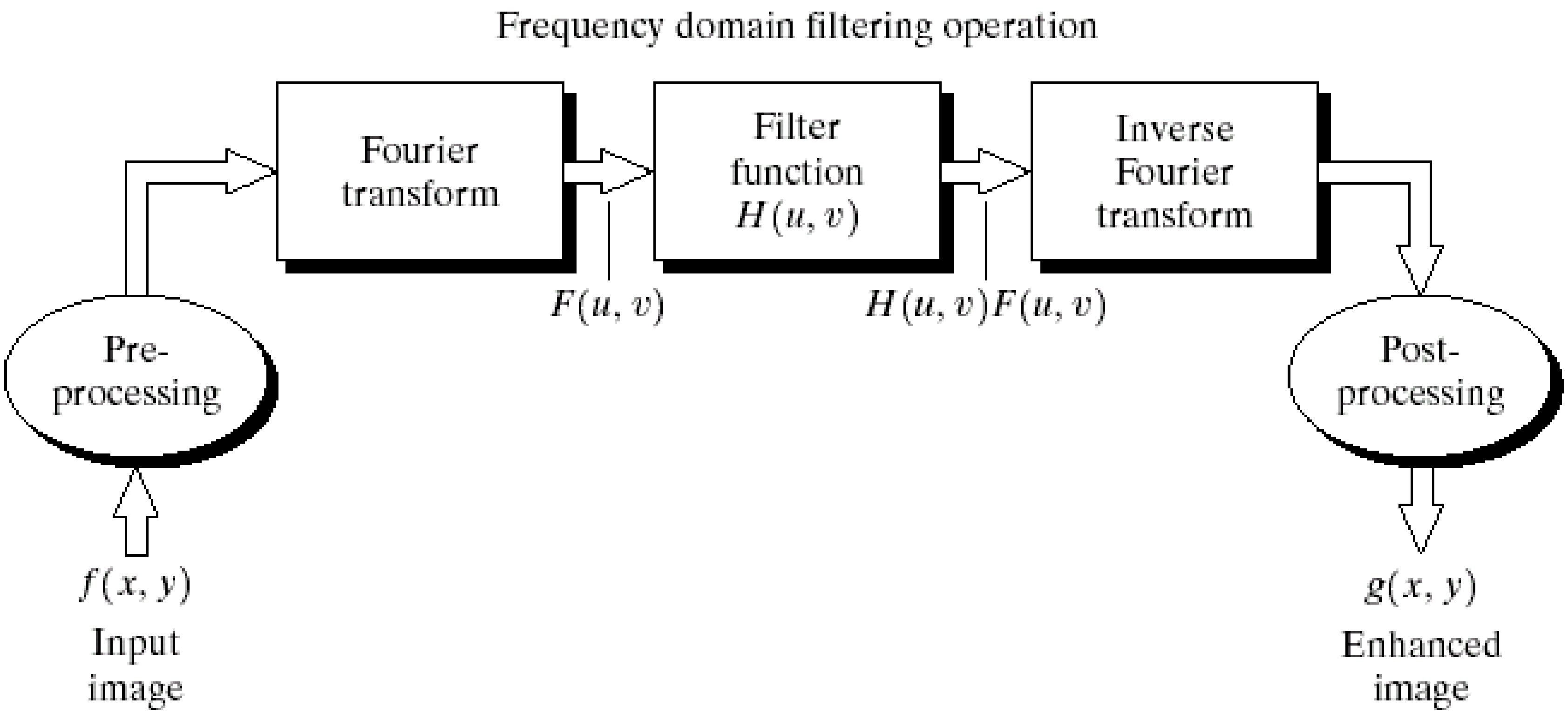


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Filtering in the Frequency Domain

•*Basic Steps for Filtering in the Frequency Domain:*

1. *Multiply the input image by $(-1)^{x+y}$ to center the transform*
2. *Compute F(u,v), the DFT of the image from (1)*
3. *Multiply F(u,v) by a filter function H(u,v)*
4. *Compute the inverse DFT of the result in (3)*
5. *Obtain the real part of the result in (4)*
6. *Multiply the result in (5) by $(-1)^{x+y}$*

Given the filter H(u,v) (filter transfer function) in the frequency domain, the Fourier transform of the output image (filtered image) is given by:

$$G(u,v) = H(u,v)F(u,v) \qquad \textit{Step (3)}$$

The filtered image g(x,y) is simply the inverse Fourier transform of G(u,v).

$$g(x,y) = \Im^{-1}\left[G(u,v)\right] \qquad \textit{Step (4)}$$

Filtering in the Frequency Domain

•Basics of Low Pass Filters in the Frequency Domain:

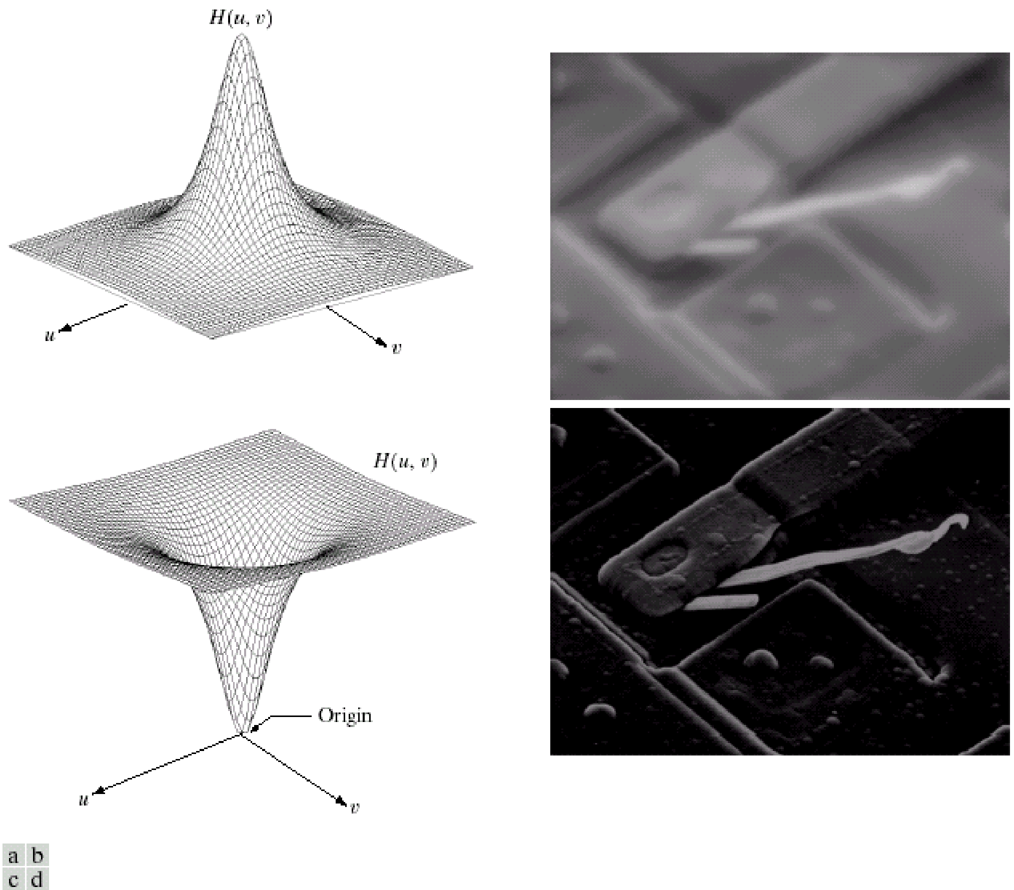


FIGURE 4.7 (a) A two-dimensional lowpass filter function. (b) Result of lowpass filtering the image in Fig. 4.4(a). (c) A two-dimensional highpass filter function. (d) Result of highpass filtering the image in Fig. 4.4(a).

•lowpass filter: A filter that attenuates high frequencies while passing the low frequencies.

•Low frequencies represent the gray-level appearance of an image over smooth areas.

•highpass filter: A filter that attenuates low frequencies while passing the high frequencies.

•High frequencies represents the details such as edges and noise.

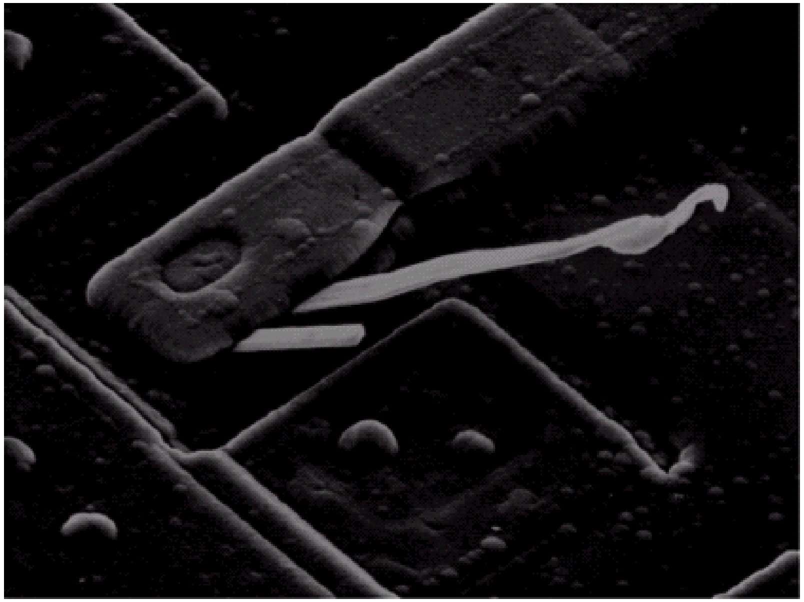
Filtering in the Frequency Domain

- *Consider the following filter transfer function:*

$$H(u,v) = \begin{cases} 0 & if \;\; (u,v) = (M/2, N/2) \\ 1 & otherwise \end{cases}$$

- *This filter will set F(0,0) to zero and leave all the other frequency components. Such a filter is called the notch filter, since it is constant function with a hole (notch) at the origin.*

FIGURE 4.6
Result of filtering the image in Fig. 4.4(a) with a notch filter that set to 0 the $F(0,0)$ term in the Fourier transform.

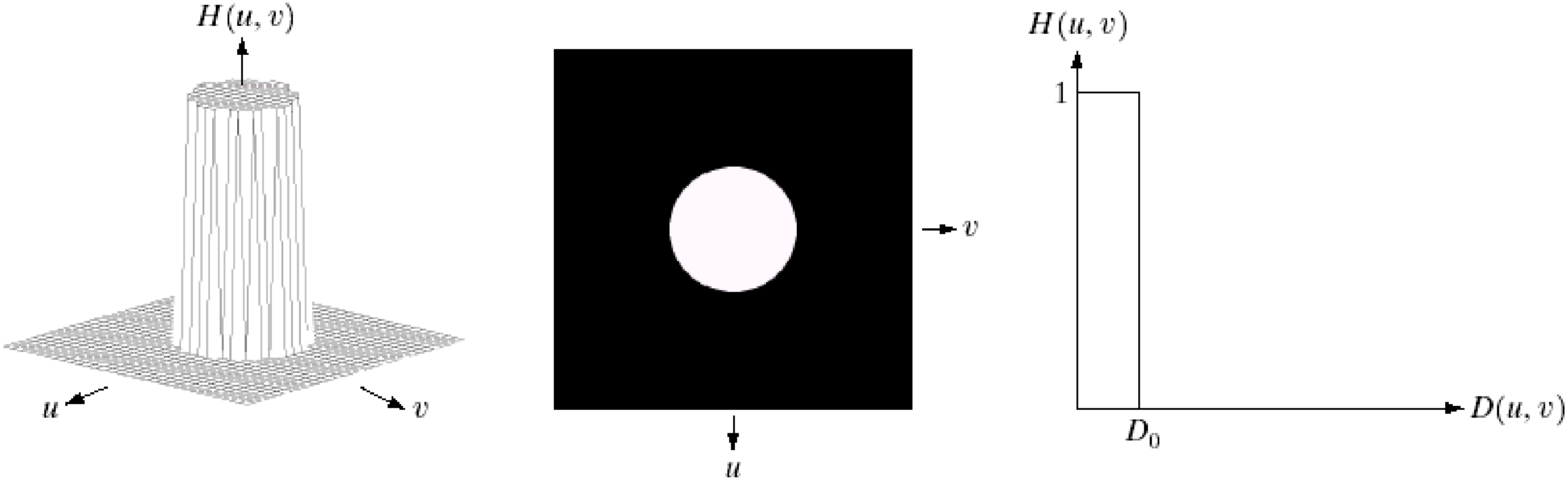


Filtering in the Frequency Domain

- ***Smoothing Frequency Domain Filters: Ideal Lowpass Filters***

$$H(u,v)=\begin{cases}1 & if\ \ D(u,v)\le D_0\\ 0 & if\ \ D(u,v)>D_0\end{cases}$$

$D(u,v)$ is the distance from the origin
D_0 is the cutoff frequency



a b c

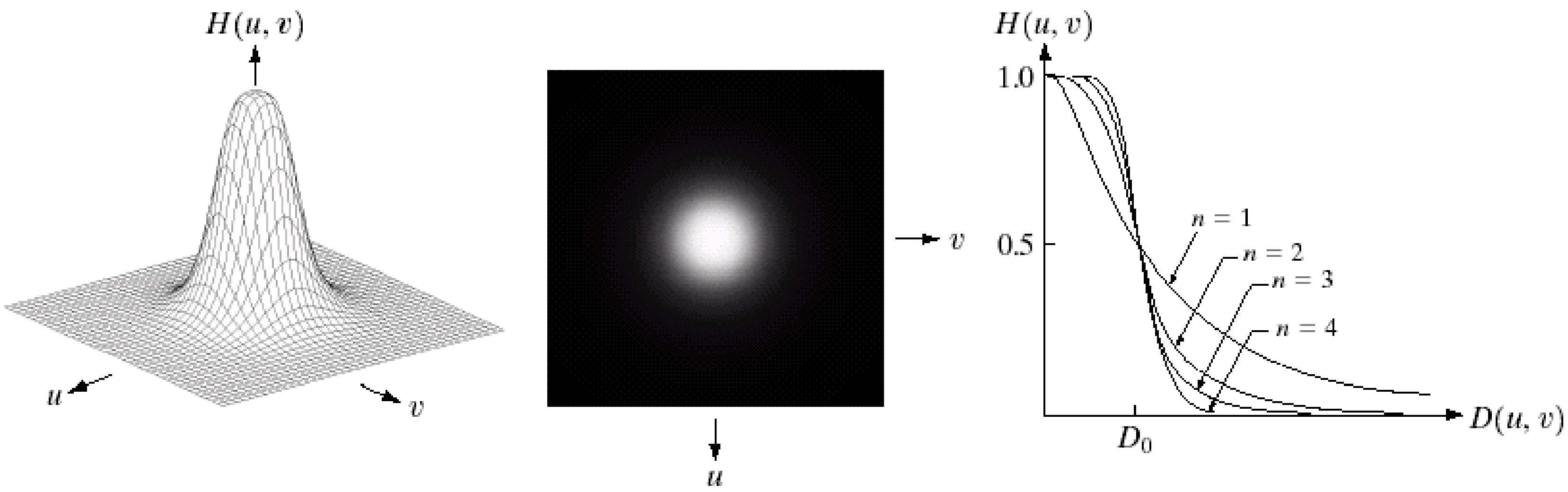
FIGURE 4.10 (a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

Filtering in the Frequency Domain

- ***Smoothing Frequency Domain Filters: Butterworth Lowpass Filters***

$$H(u,v) = \frac{1}{1+[D(u,v)/D_0]^{2n}}$$

D(u,v) is the distance from the origin
D_0 is the cutoff frequency.
n is the order of the filter



a b c

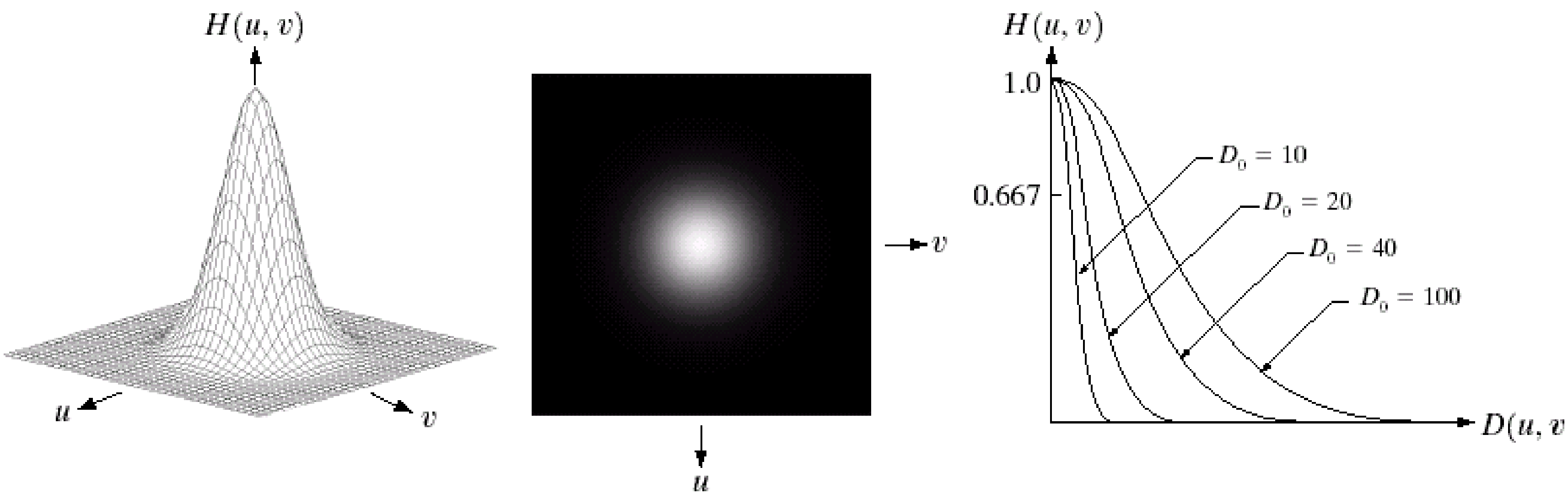
FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Filtering in the Frequency Domain

- ***Smoothing Frequency Domain Filters: Gaussian Lowpass Filters***

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

$D(u,v)$ is the distance from the origin
D_0 is the cutoff frequency.

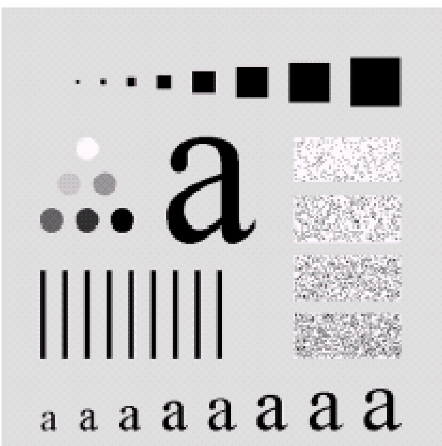


a b c

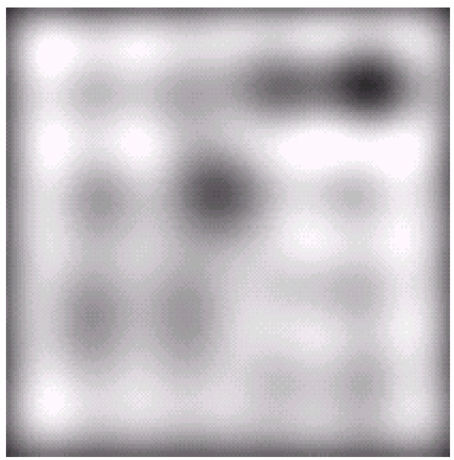
FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.

Filtering in the Frequency Domain

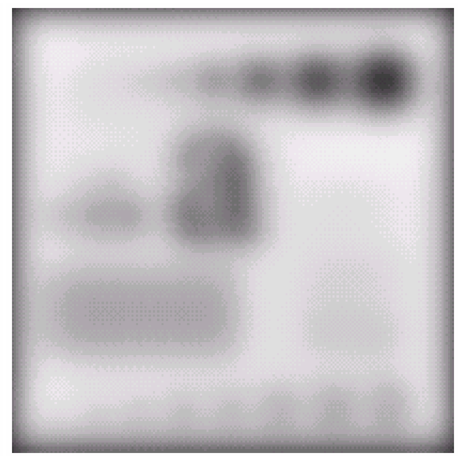
•*Comparison of Ideal, Butterworth and Gaussian Lowpass Filters having the same radii (cutoff frequency) value.*



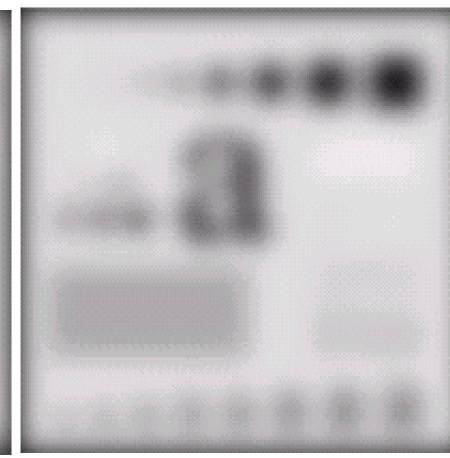
Original image



Ideal LP Filtered



Butterworth LP Filtered



Gaussian LP Filtered

Smoothing Frequency-Domain Filters

- *The high-frequency components are: edges and sharp transitions such as noise.*

- *Smoothing/blurring can be achieved by attenuating a specified range of high-frequency components in the frequency domain.*

- *Given the Fourier transformed image **F(u)** , the filtered image **G(u)** can be obtained by:*

$$G(u,v) = H(u,v)F(u,v)$$

- *Where **H(u)** is the filter transfer function.*

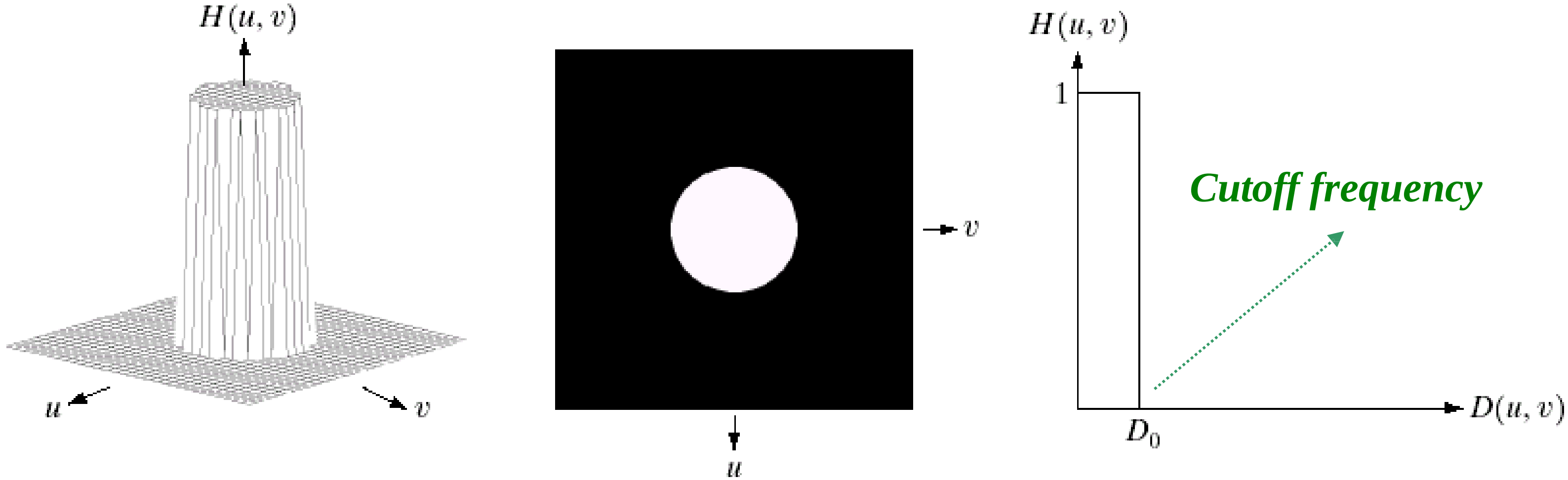
- *Smoothing can be achieved by lowpass filters. We will consider only 3 types of lowpass filters:*
    - *Ideal Lowpass filters,*
    - *Butterworth Lowpass filters,*
    - *Gaussian Lowpass filters*

Smoothing Frequency-Domain Filters

•**Ideal Lowpass Filter (ILPF):** ***Simply cuts off all the high frequencies higher than the specified cutoff frequency. The filter transfer function:***

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

D(u,v) ***is the distance from the origin***
D_0 is the cutoff frequency



a b c

FIGURE 4.10 (a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

Smoothing Frequency-Domain Filters

- **Cutoff Frequency of an Ideal Lowpass Filter:** ***One way of specifying the cutoff frequency is to compute circles enclosing specified amounts of total image power.***
- ***Calculating P_T which is the total power of the transformed image:***

$$P_T = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} P(u,v) \qquad u = 0, 1, \ldots, M-1 \quad , v = 0, 1, \ldots, N-1$$

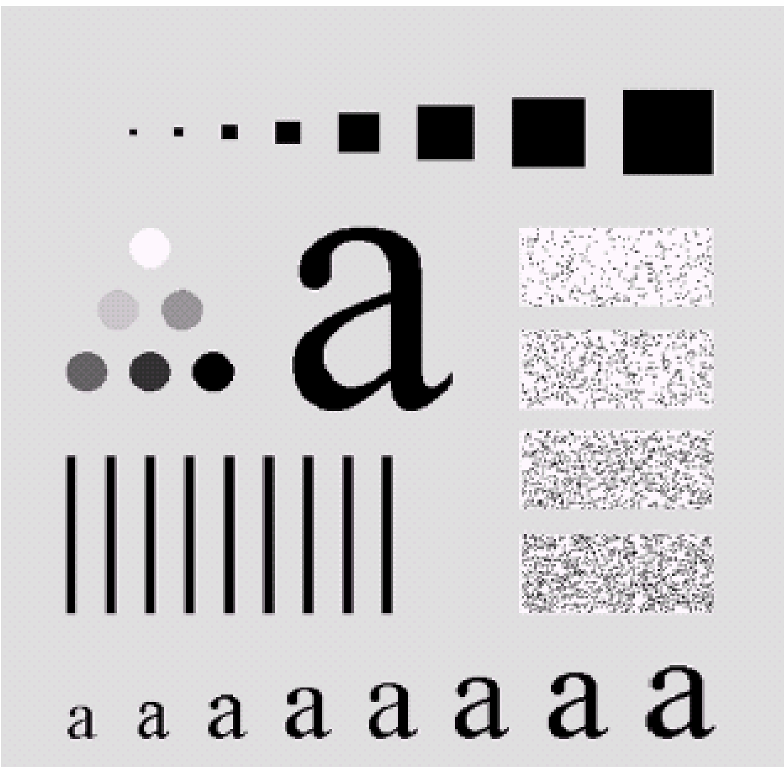
- ***The cutoff frequency D_o can be determined by specifying the α percent of the total power enclosed by a circle centered at the origin. The radius of the circle is the cutoff frequency D_o.***

$$\alpha = 100 \left[\sum_u \sum_v P(u,v) / P_T \right] \qquad u \le radius(D_o) \quad , v \le radius(D_o)$$

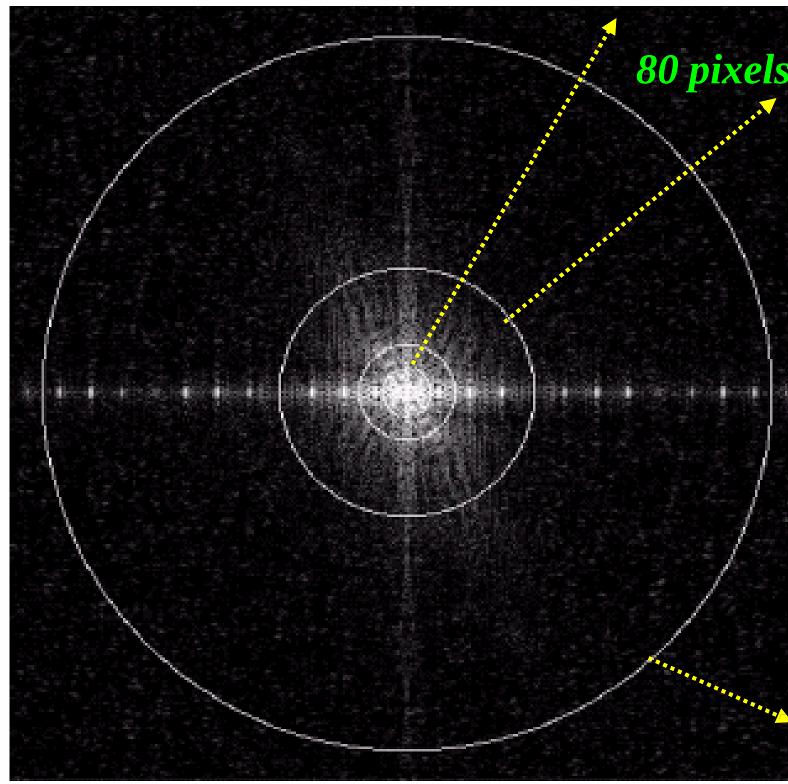
Smoothing Frequency-Domain Filters

- **Cutoff Frequency of an Ideal Lowpass Filter:**

Original Image



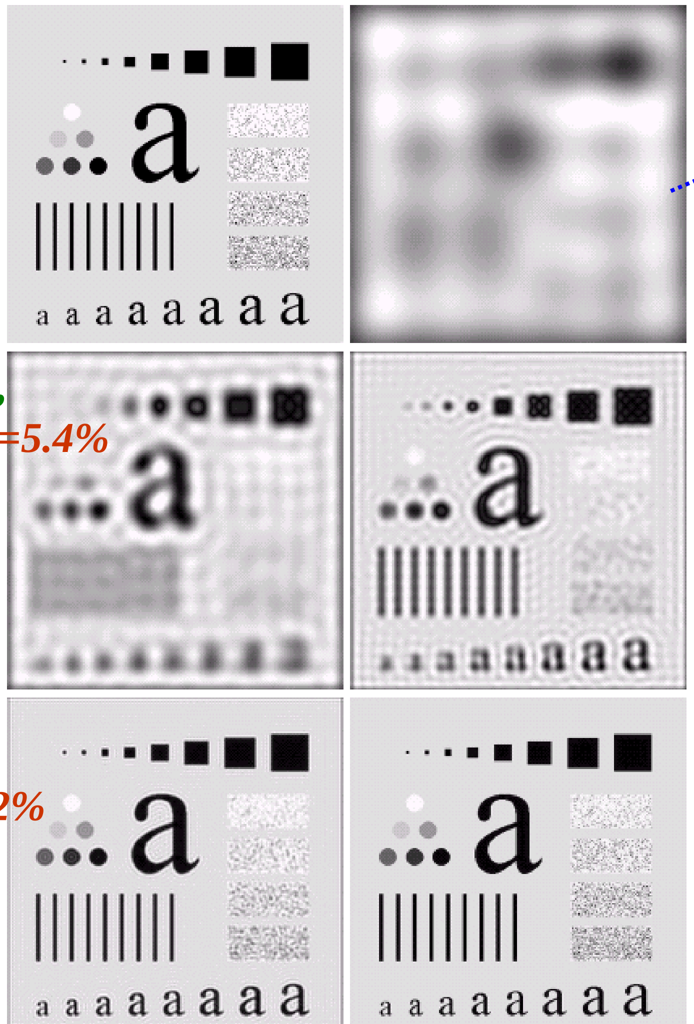
Fourier Spectrum



Smoothing Frequency-Domain Filters

- **Cutoff Frequency of an Ideal Lowpass Filter:**

Original Image



ILPF with cutoff=5, removed power=8%

Blurring effect is the result of LPIF

cutoff=15,
r.d. power=5.4%

ILPF with cutoff=30, removed power=3.6%

Ringing effect is the problem of LPIF

cutoff=80,
r.d. power=2%

ILPF with cutoff=230, removed power=0.5%

Smoothing Frequency-Domain Filters

- **Blurring and Ringing properties of *ILPF*:**
- ***The blurring and ringing properties of the ILPF can be explained by the help of the convolution theorem:***
- ***Given the Fourier transformed input image* F(u) *and output image* G(u) *and the filter transfer function* H(u),**

$$G(u,v) = H(u,v)F(u,v)$$

- ***The corresponding process in the spatial domain with regard to the convolution theorem is given by:***

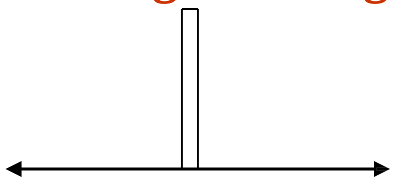
$$g(x,y) = h(x,y) * f(x,y)$$

- ***h(x,y) in the spatial domain, is the inverse Fourier transform of the filter transfer function H(u,v).***

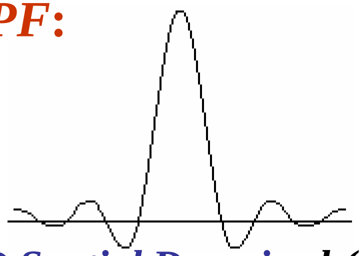
- ***The spatial filter h(x,y) has two major characteristics:***
  - ***dominant component at the origin***
  - ***Concentric circular components about center component.***

Smoothing Frequency-Domain Filters

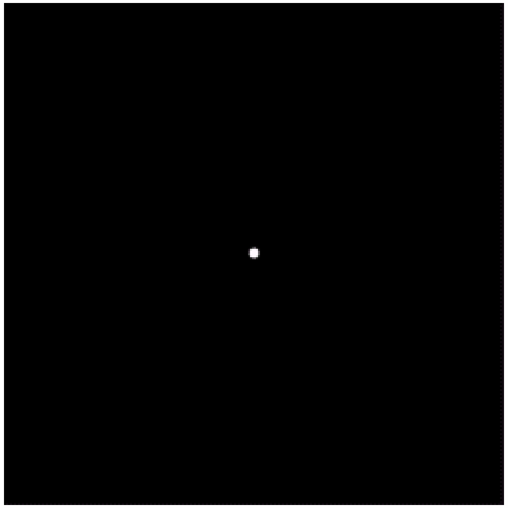
- **Blurring and Ringing properties of *ILPF*:**



1D Frequency Domain- H(u)

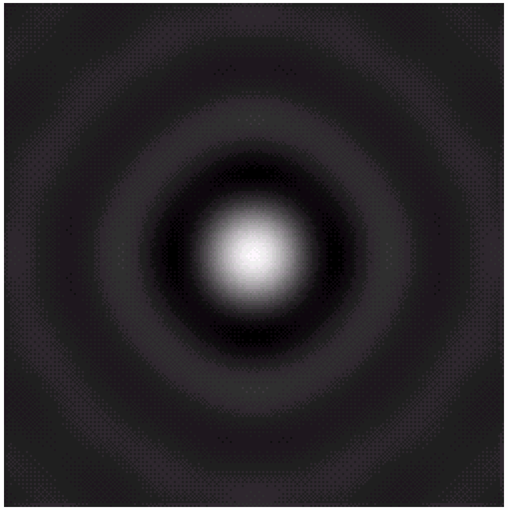


1D Spatial Domain- h(x)



2D Frequency Domain- H(u,v)

inverse DFT



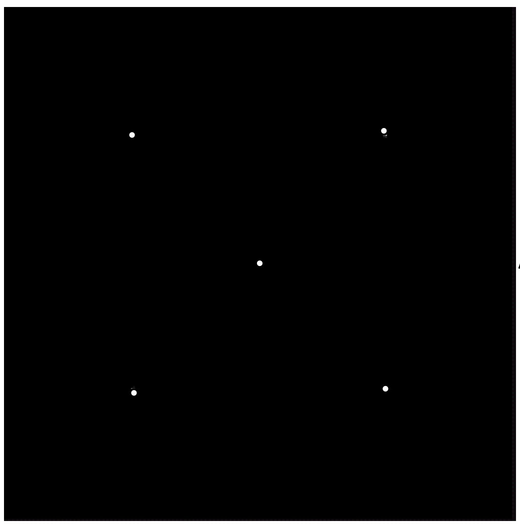
2D Spatial Domain – h(x,y)

- ***The spatial domain filters center component is responsible for blurring.***
- ***The circular components are responsible for the ringing artifacts.***

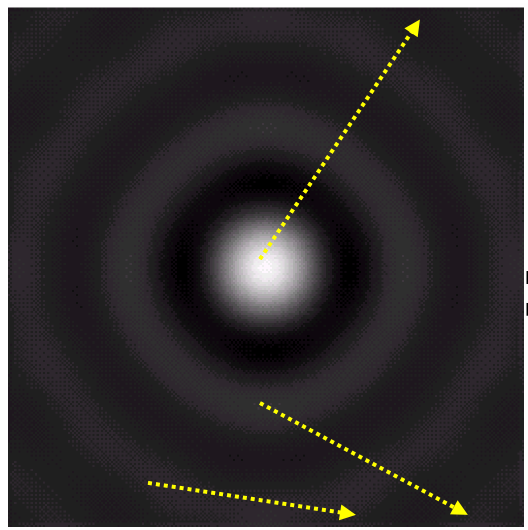
Smoothing Frequency-Domain Filters

- **Blurring and Ringing properties of *ILPF*:** ***Lets consider the following convolution in the spatial domain:***

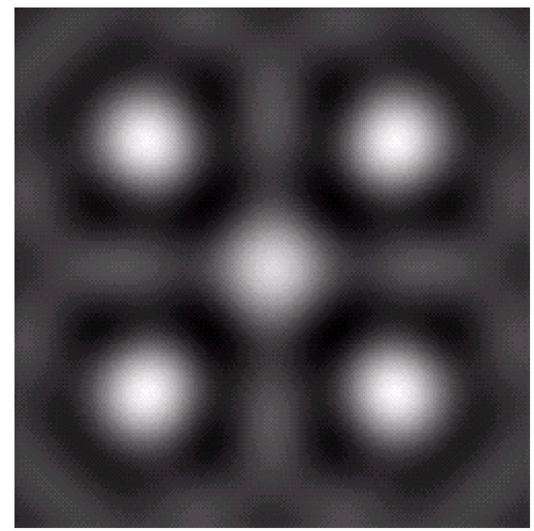
Central components: causes blurring



*



=



circular components: causes ringing effect

Input image - f(x,y)

Spatial Filter – h(x,y)

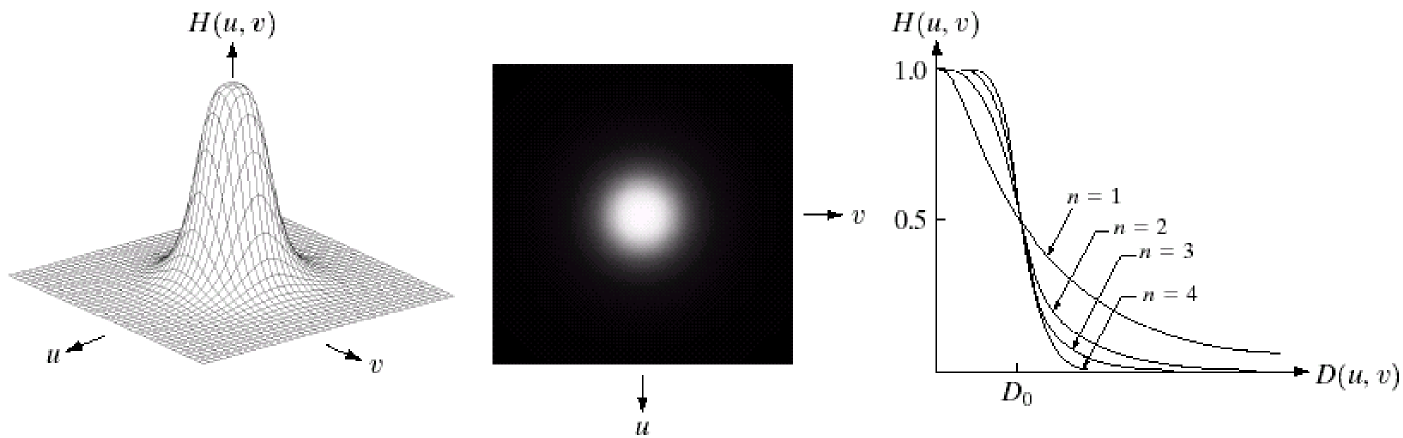
filtered image – g(x,y)

Smoothing Frequency-Domain Filters

•**Butterworth Lowpass Filter (BLPF):** *The transfer function of BLPF of order n and with a specified cutoff frequency is denoted by the following filter transfer function:*

$$H(u,v) = \frac{1}{1+[D(u,v)/D_0]^{2n}}$$

D(u,v) *is the distance from the origin*
D_0 *is the cutoff frequency.*
n *is the order of the filter*

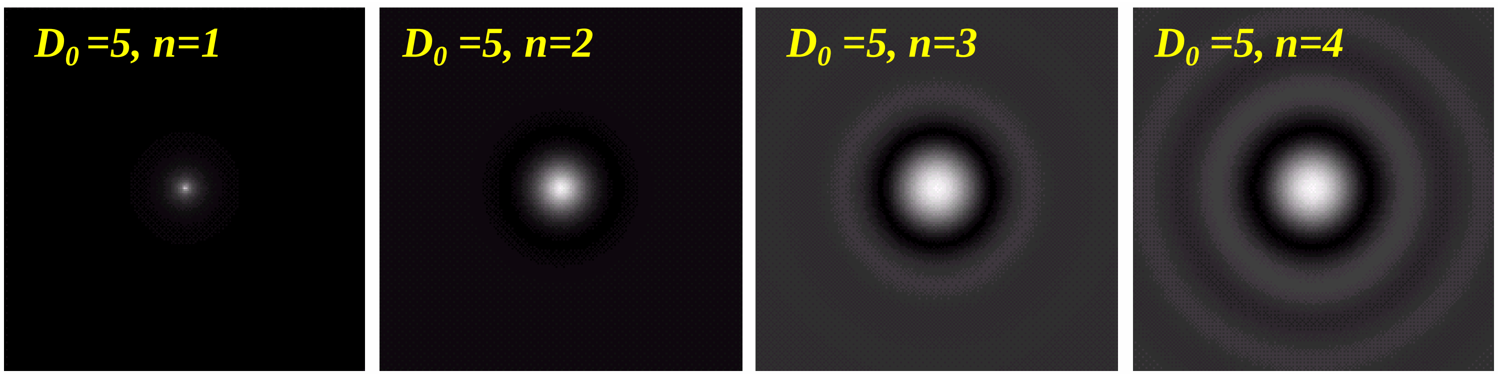


a b c

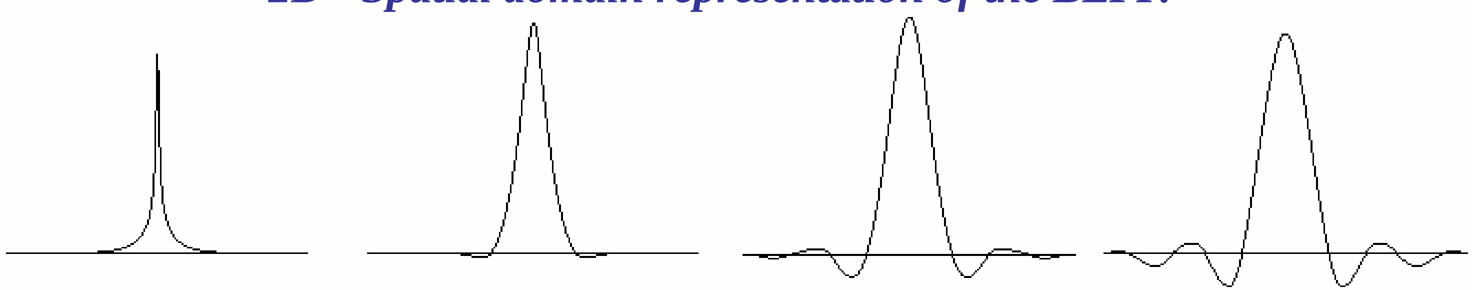
FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Smoothing Frequency-Domain Filters

- **Butterworth Lowpass Filter (BLPF):**
- ***The BLPF with order of 1 does not have any ringing artifact.***
- ***BLPF with orders 2 or more shows increasing ringing effects as the order increases.***



2D - Spatial domain representation of the BLPF.



1D - Profile through the centre of the Spatial domain filters.

Smoothing Frequency-Domain Filters

- **Butterworth Lowpass Filter (BLPF):**

Original Image

BLPF with cutoff=5, order=2

cutoff=15, order=2

BLPF with cutoff=30, order=2

cutoff=80, order=2

BLPF with cutoff=230, order=2

Smoothing Frequency-Domain Filters

•**Gaussian Lowpass Filter (GLPF):** ***The transfer function of GLPF is given as follows:***

$$\boxed{H(u,v) = e^{-D^2(u,v)/2D_0^2}}$$

***D(u,v)** is the distance from the origin*
***D_0** is the cutoff frequency.*

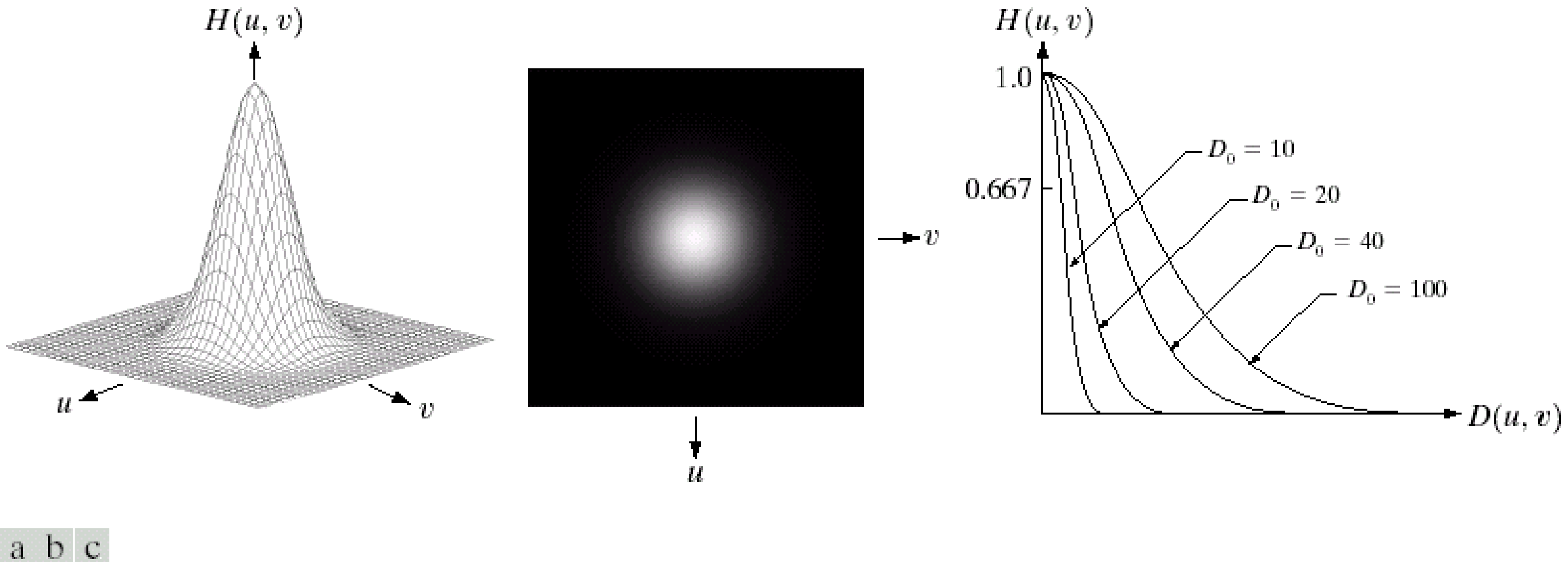
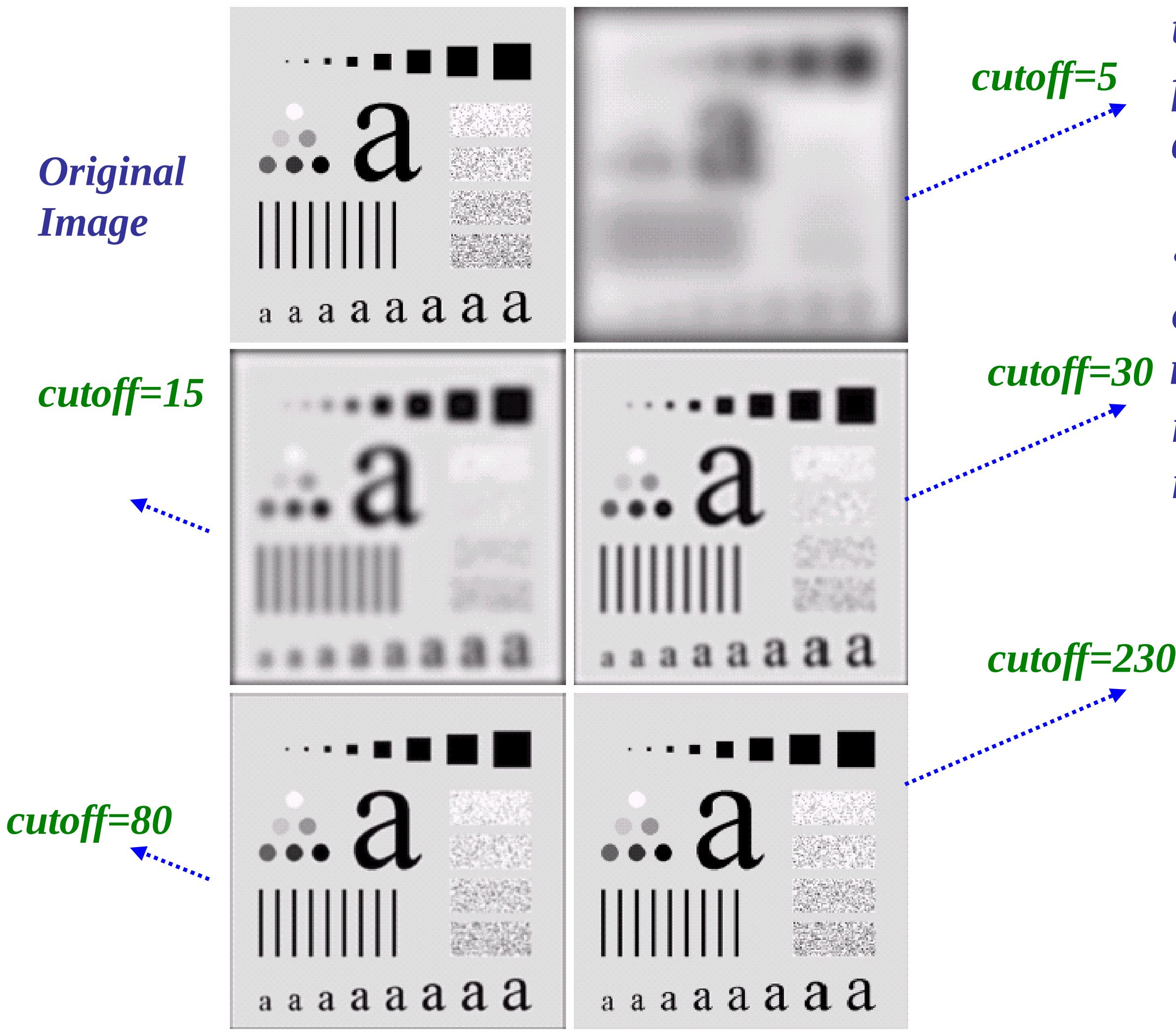


FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.

Smoothing Frequency-Domain Filters



- **(GLPF):** *The inverse Fourier transform of the Gaussian Lowpass filter is also Gaussian in the Spatial domain.*

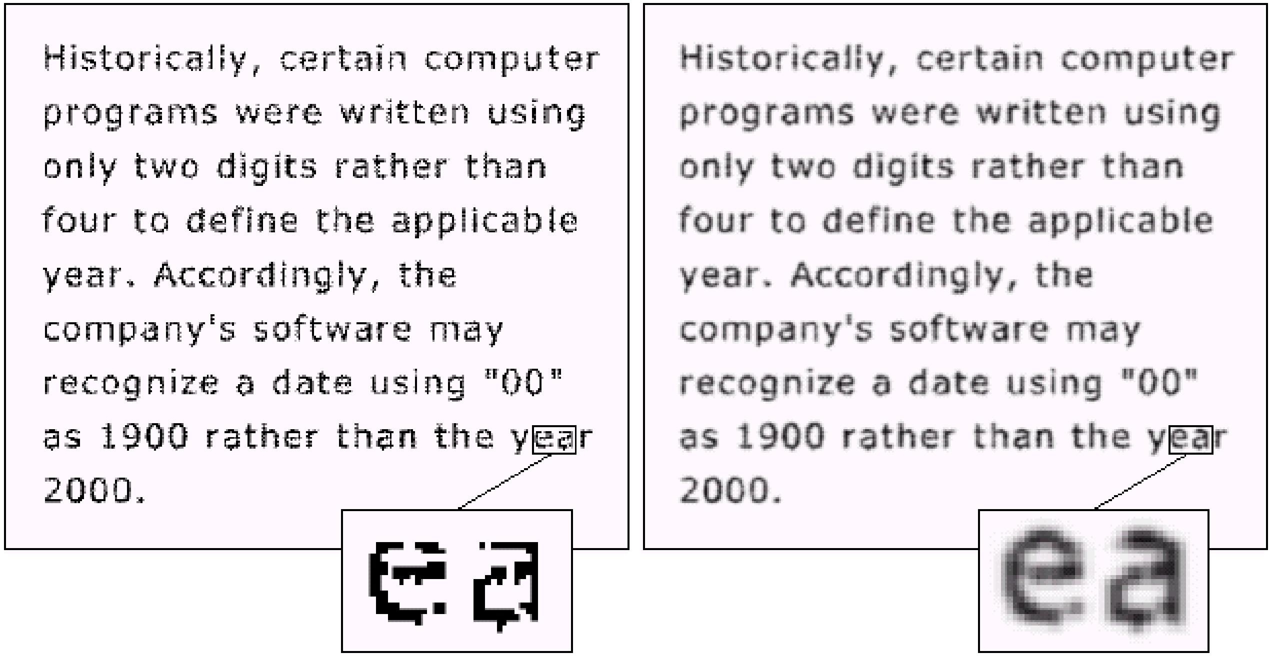
- *Therefore there is no ringing effect of the GLPF. Ringing artifacts are not acceptable in fields like medical imaging. Hence use Gaussian instead of the ILPF/BLPF.*

Smoothing Frequency-Domain Filters

- **Gaussian Lowpass Filter (GLPF):** ***Refer to the improvement in the following example.***

a b

FIGURE 4.19
(a) Sample text of poor resolution (note broken characters in magnified view). (b) Result of filtering with a GLPF (broken character segments were joined).



Smoothing Frequency-Domain Filters

- **Gaussian Lowpass Filter (GLPF):** ***The following example shows a lady younger. How? By using GLPF!***

a b c

FIGURE 4.20 (a) Original image (1028 × 732 pixels). (b) Result of filtering with a GLPF with $D_0 = 100$. (c) Result of filtering with a GLPF with $D_0 = 80$. Note reduction in skin fine lines in the magnified sections of (b) and (c).

Sharpening Frequency-Domain Filters

- *The high-frequency components are: edges and sharp transitions such as noise.*

- *Sharpening can be achieved by highpass filtering process, which attenuates low frequency components without disturbing the high-frequency information in the frequency domain.*

- *The filter transfer function, $H_{hp}(u,v)$, of a highpass filter is given by:*

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

- *Where $H_{lp}(u,v)$, is the transfer function of the corresponding lowpass filter.*

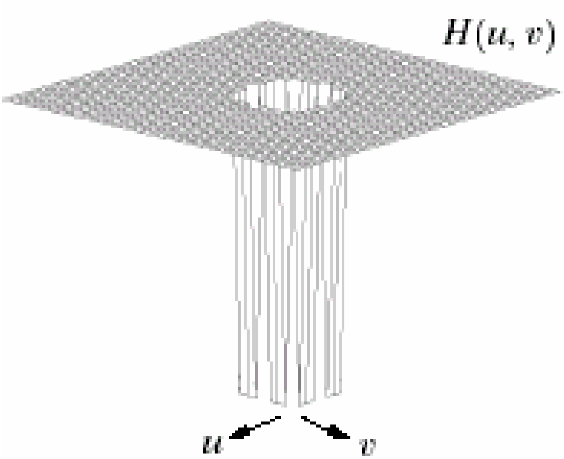
- *We will consider only 3 types of sharpening highpass filters :*
  - *Ideal Highpass filters,*
  - *Butterworth Highpass filters,*
  - *Gaussian Highpass filters*

Sharpening Frequency-Domain Filters

•**Ideal Highpass Filter (IHPF):** ***Simply cuts off all the low frequencies lower than the specified cutoff frequency. The filter transfer function:***

$$H(u,v)=\begin{cases}0 & \text{if } D(u,v)\le D_0\\ 1 & \text{if } D(u,v)> D_0\end{cases}$$

$D(u,v)$ is the distance from the origin
D_0 is the cutoff frequency



Perspective plot of IHPF

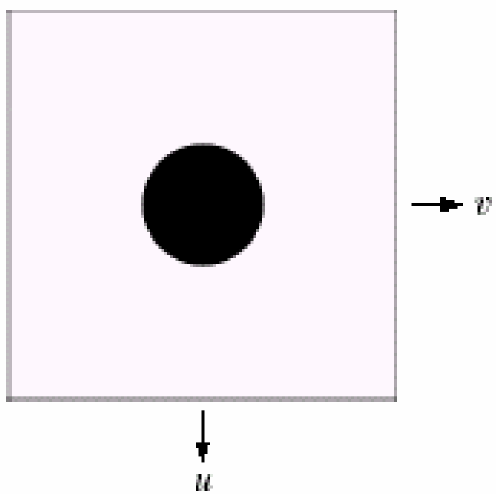
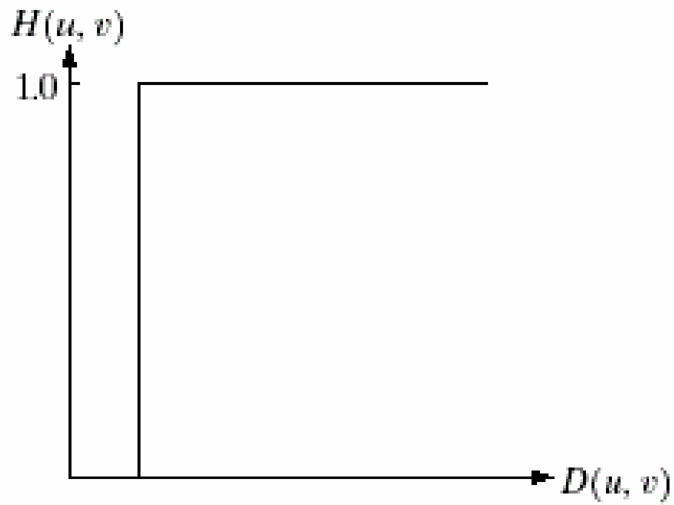


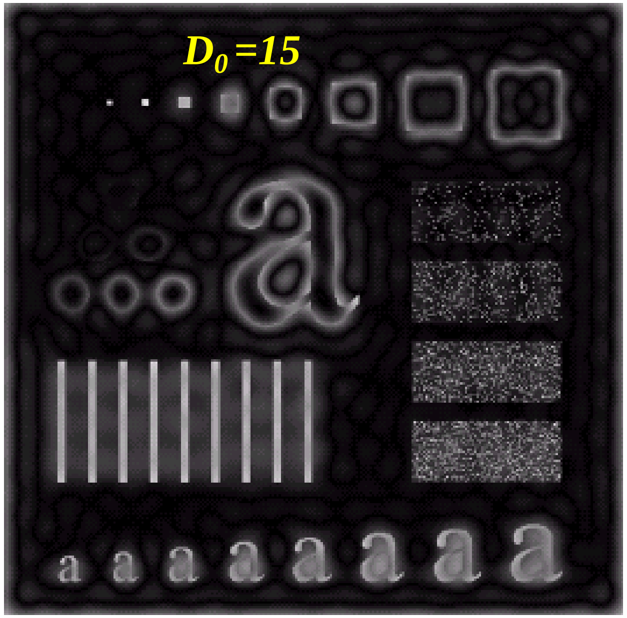
Image representation of IHPF

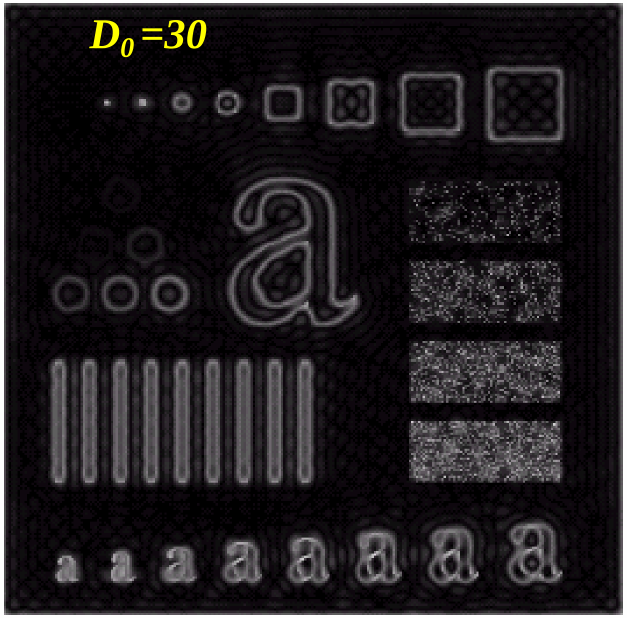


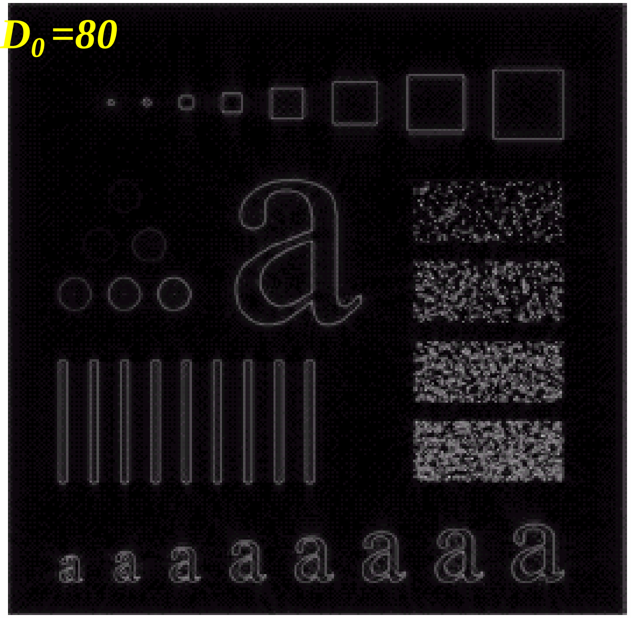
Cross section of IHPF

Sharpening Frequency-Domain Filters

•**Ideal Highpass Filter (IHPF):** ***The ringing artifacts occur at low cutoff frequencies***





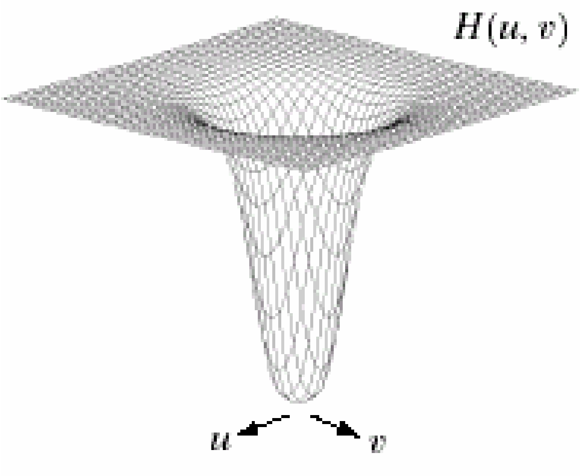


Sharpening Frequency-Domain Filters

•**Butterworth Highpass Filter (BHPF):** *The transfer function of BHPF of order n and with a specified cutoff frequency is given by:*

$$\boxed{H(u,v) = \frac{1}{1+[D_0 / D(u,v)]^{2n}}}$$

D(u,v) *is the distance from the origin*
D_0 *is the cutoff frequency.*
n *is the order of the filter*



Perspective plot of BHPF

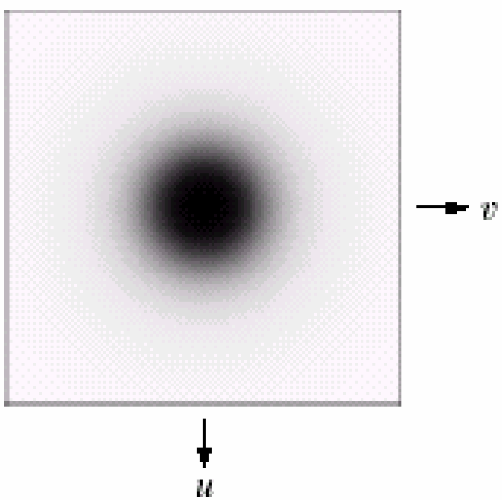
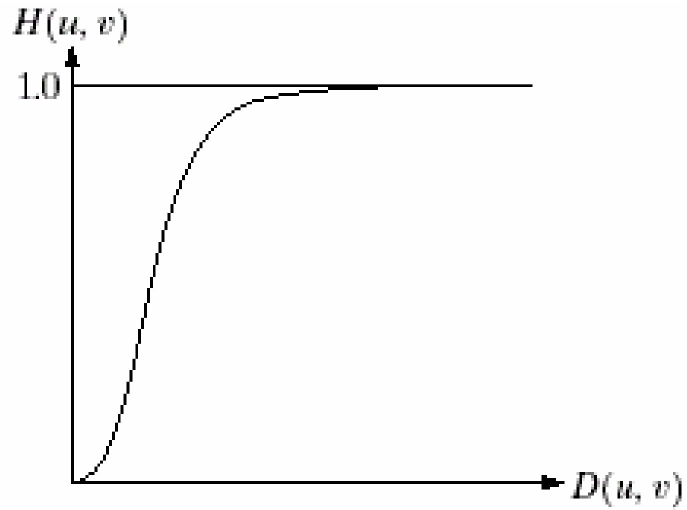


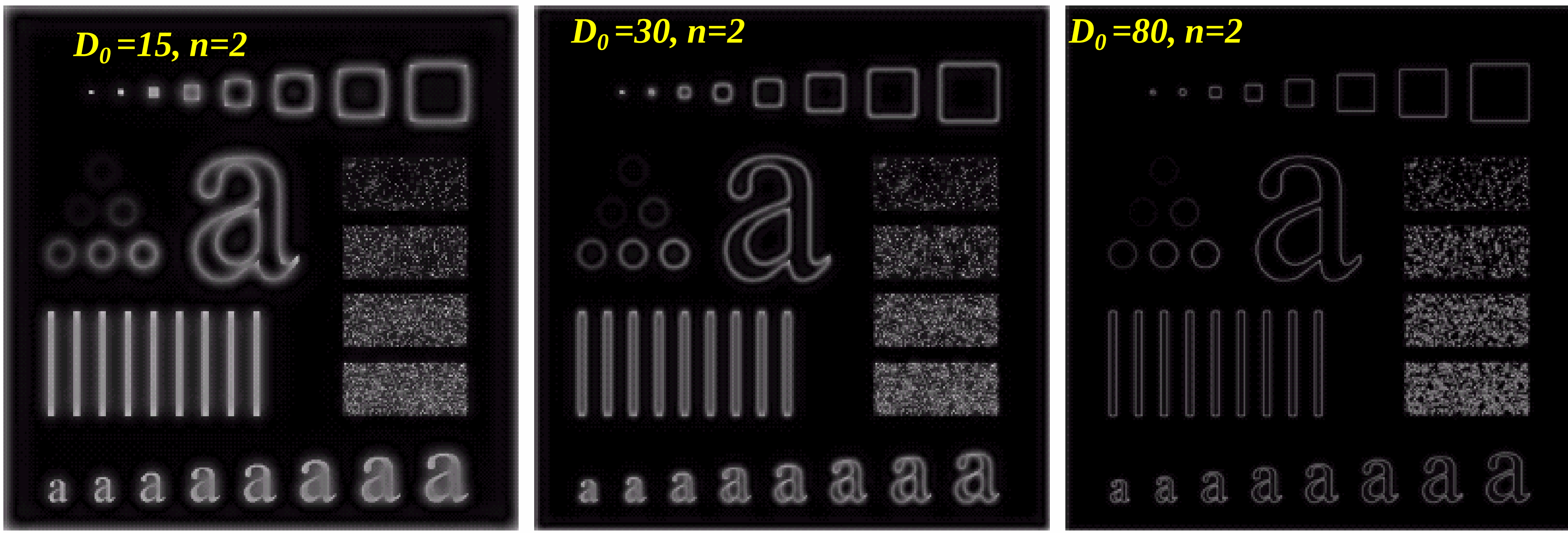
Image representation of BHPF



Cross section of BHPF

Sharpening Frequency-Domain Filters

•**Butterworth Highpass Filter (BHPF):** ***Smoother results are obtained in BHPF when compared IHPF. There is almost no ringing artifacts.***

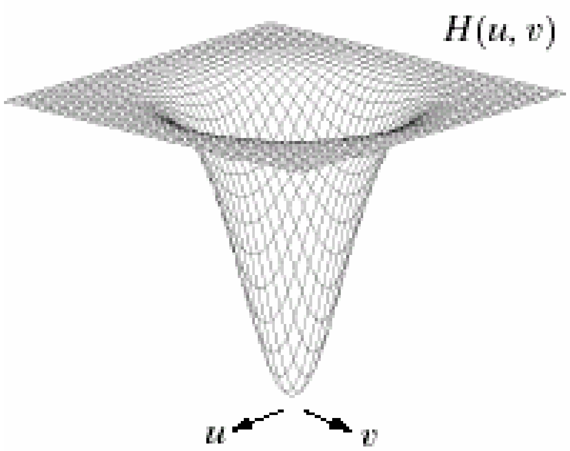


Sharpening Frequency-Domain Filters

•**Gaussian Highpass Filter (GHPF):** ***The transfer function of GHPF is given by:***

$$\boxed{H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}}$$

***D(u,v)** is the distance from the origin*
***D_0** is the cutoff frequency.*



Perspective plot of BHPF

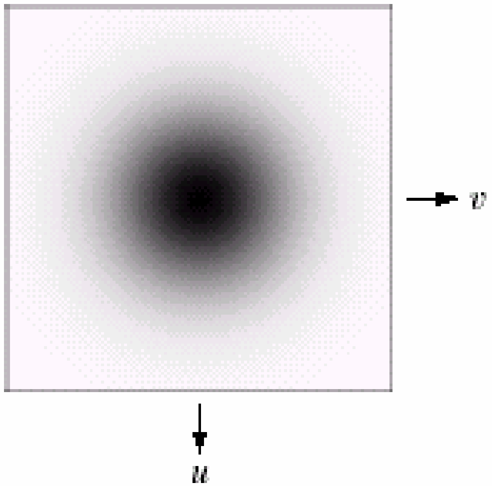
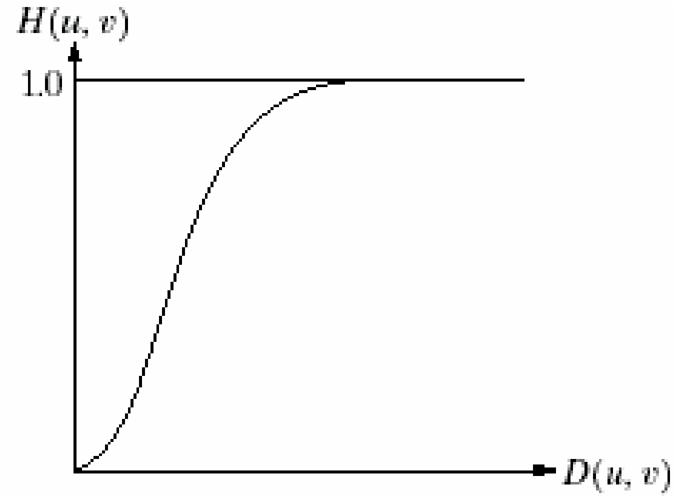


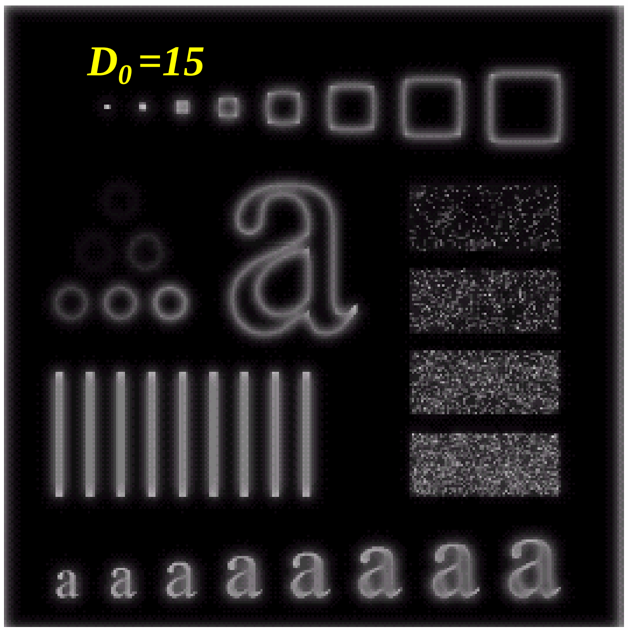
Image representation of BHPF

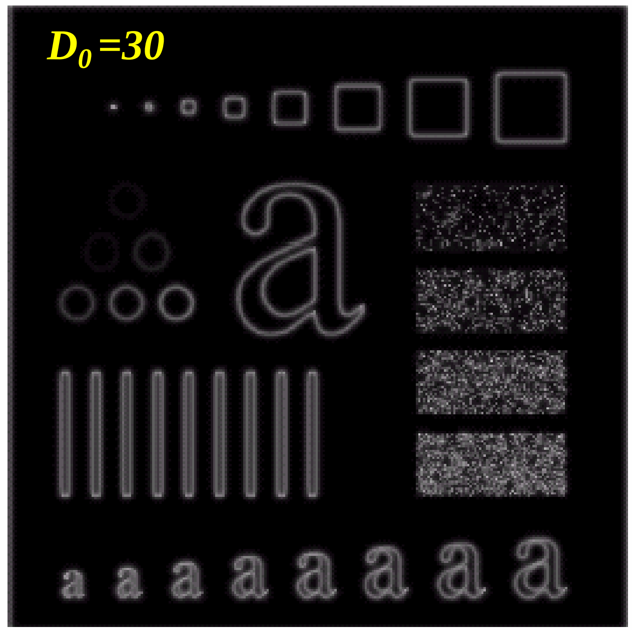


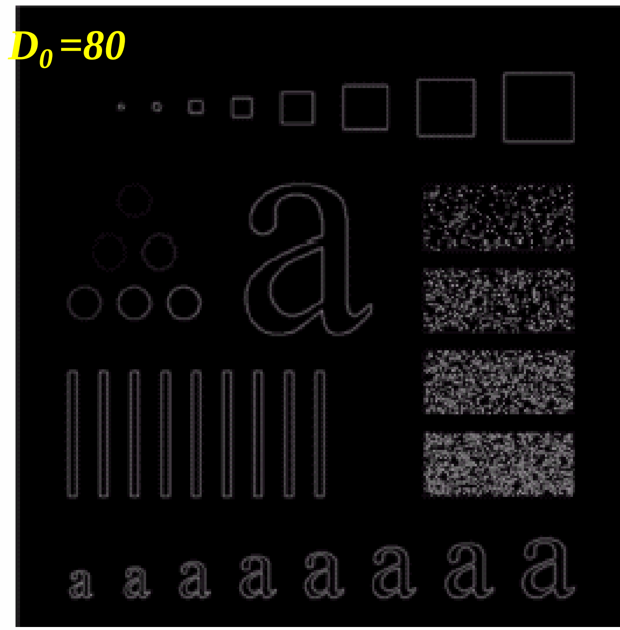
Cross section of BHPF

Sharpening Frequency-Domain Filters

- **Gaussian Highpass Filter (GHPF):** ***Smoother results are obtained in GHPF when compared BHPF. There is absolutely no ringing artifacts.***

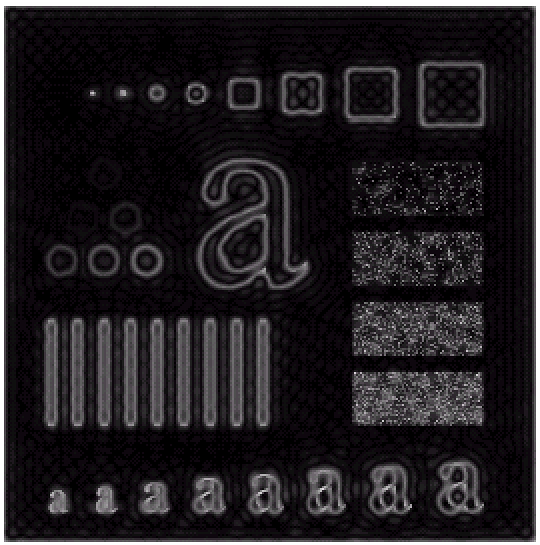




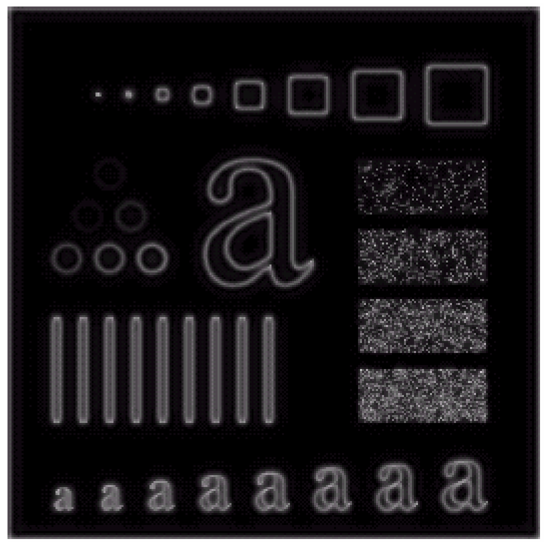


Sharpening Frequency-Domain Filters

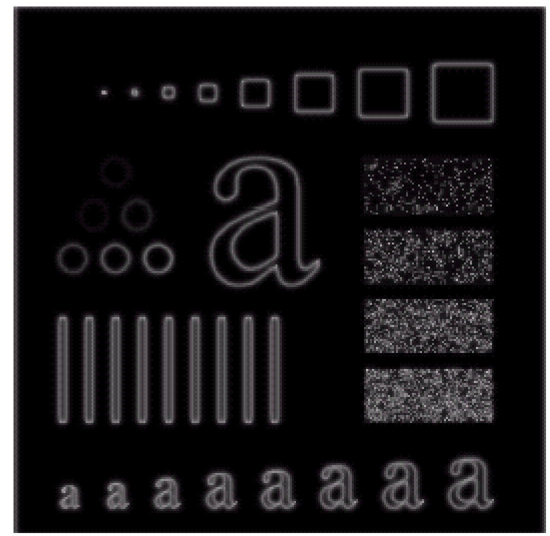
- **Comparison of Ideal, Butterworth and Gaussian High pass Filters** *($D_0 = 30$)*



Ideal High pass Filter



Butterworth High pass Filter



Gaussian High pass Filter

The distinction and links between spatial and frequency filtering

- If the size of spatial and frequency filters is same, then the computation burden in spatial domain is larger than in frequency domain;

- However, whenever possible, it makes more sense to filter in the spatial domain using small filter masks

- Filtering in frequency is more intuitive. We can specify filters in the frequency, take their inverse transform, and use the resulting filter in spatial domain as a guide for constructing smaller spatial filter masks

Laplacian in Frequency Domain

$$\Im\left[\frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}\right] = -(u^2 + v^2)F(u,v)$$

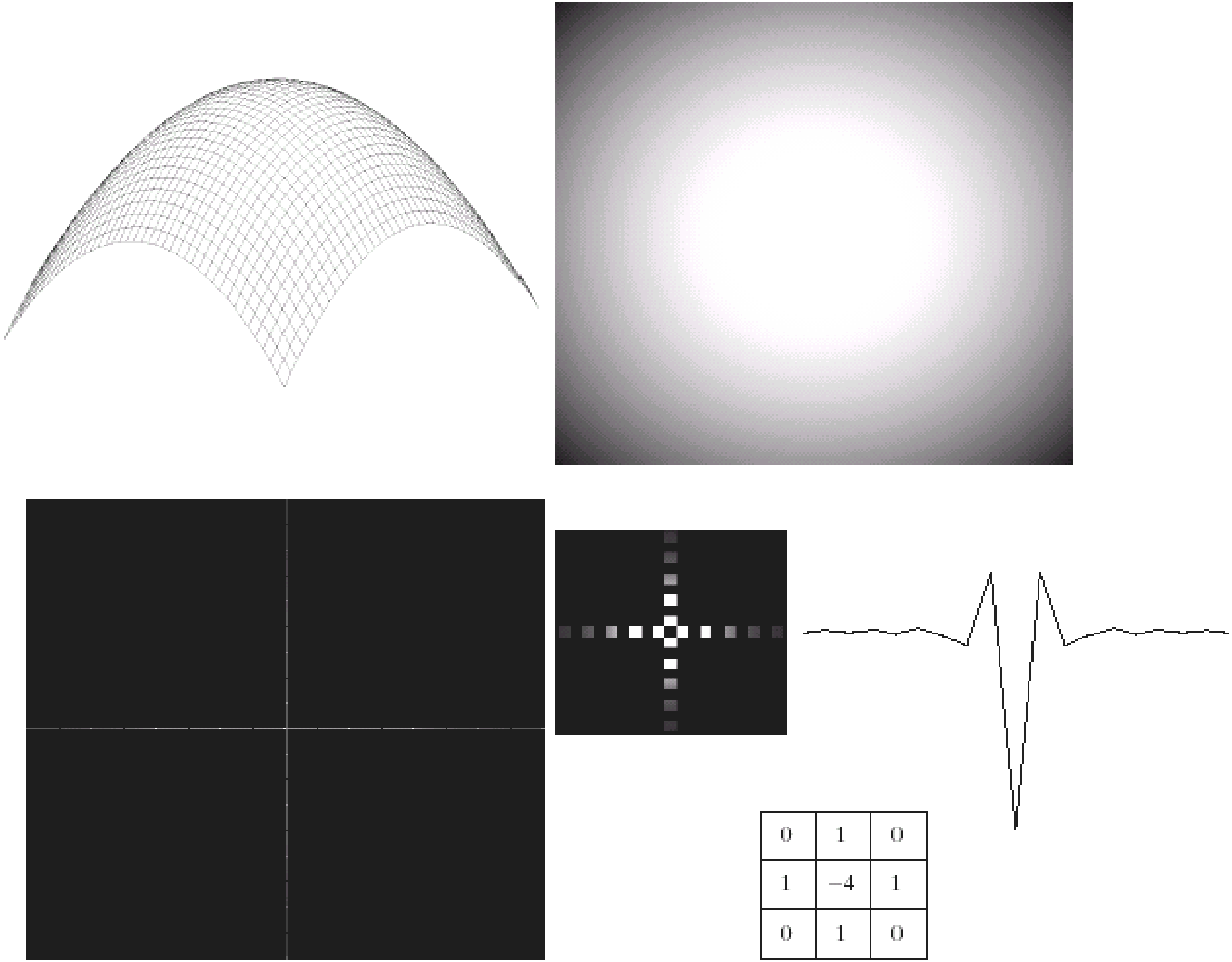
$$H_1(u,v) = -(u^2 + v^2)$$

Frequency domain

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Spatial domain

Laplacian operator



0	1	0
1	–4	1
0	1	0

a b
c d e
f

FIGURE 4.27 (a) 3-D plot of Laplacian in the frequency domain. (b) Image representation of (a). (c) Laplacian in the spatial domain obtained from the inverse DFT of (b). (d) Zoomed section of the origin of (c). (e) Gray-level profile through the center of (d). (f) Laplacian mask used in Section 3.7.

Subtract Laplacian from the Original Image to Enhance It

enhanced image — Original image — Laplacian output

Spatial domain $$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

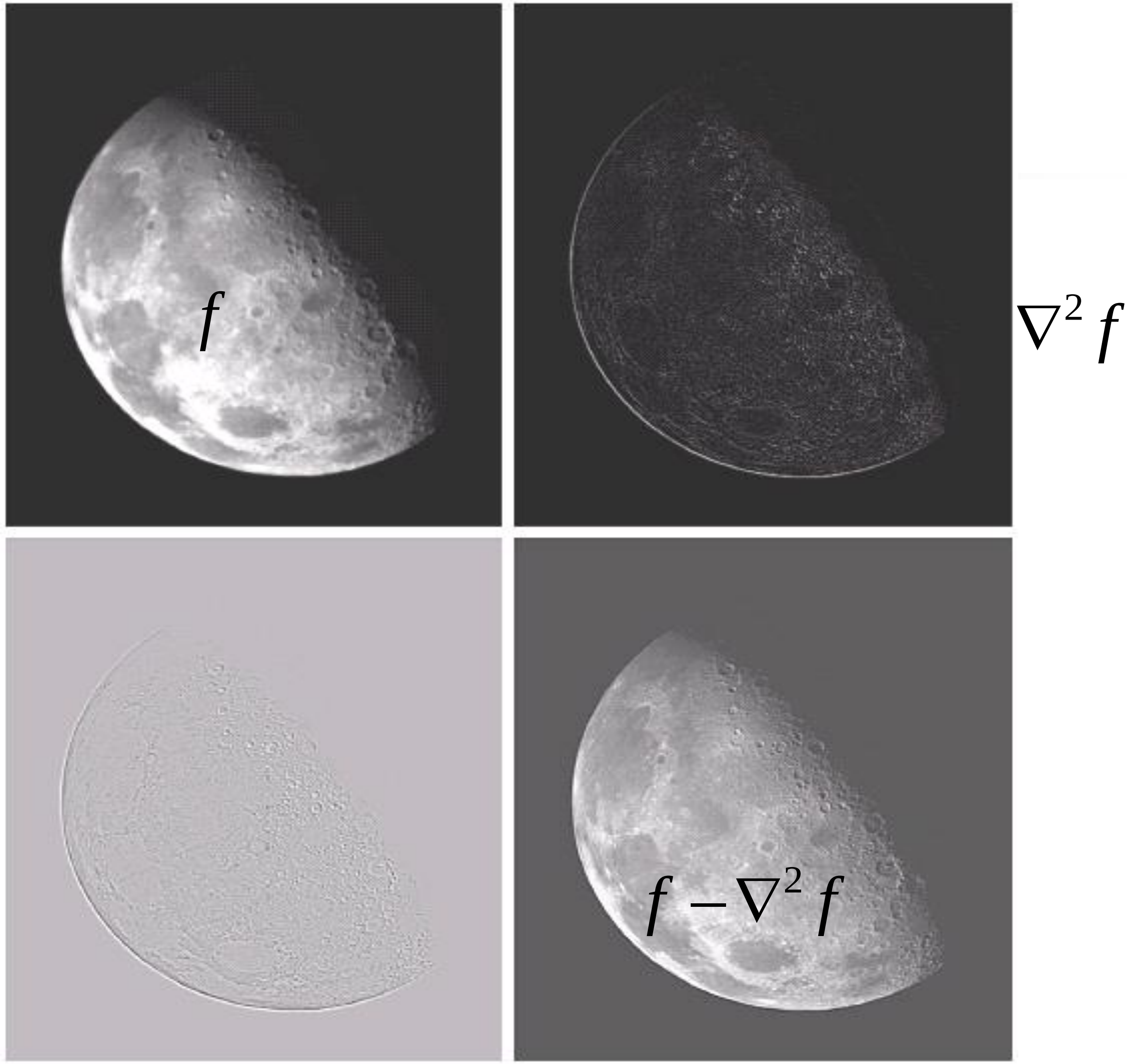
Frequency domain $$G(u,v) = F(u,v) + (u^2 + v^2)F(u,v)$$

new operator $$H_2(u,v) = 1 + (u^2 + v^2) = 1 - H_1(u,v)$$

Laplacian

a b
c d

FIGURE 4.28
(a) Image of the North Pole of the moon.
(b) Laplacian filtered image.
(c) Laplacian image scaled.
(d) Image enhanced by using Eq. (4.4-12).
(Original image courtesy of NASA.)

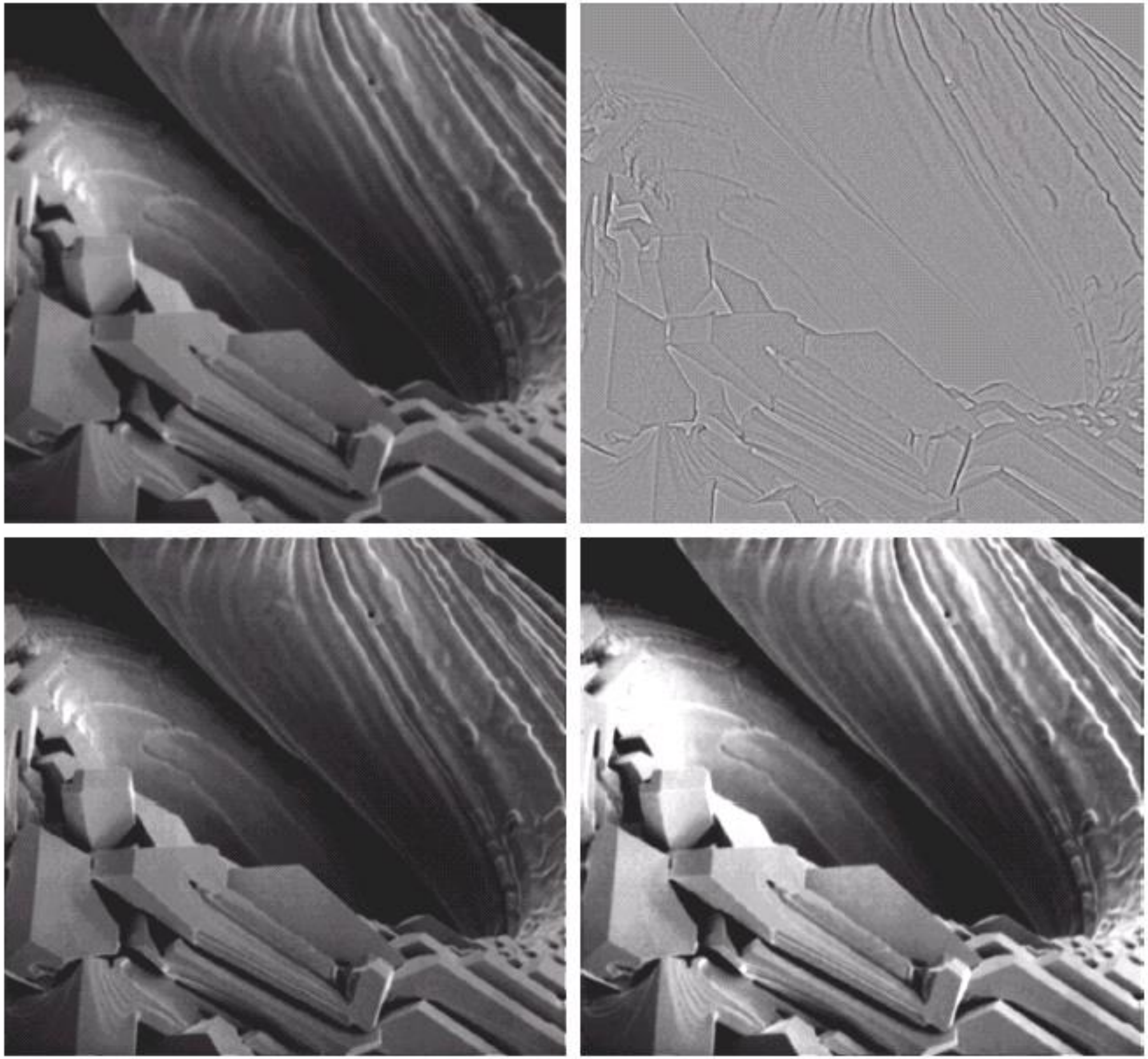


Unsharp Masking, High-boost Filtering

- Unsharp masking: $f_{hp}(x,y) = f(x,y) - f_{lp}(x,y)$
- $H_{hp}(u,v)=1 - H_{lp}(u,v)$

- High-boost filtering: → One more parameter to adjust the enhancement

  $f_{hb}(x, y) = Af(x, y) - f_{lp}(x, y)$
- $f_{hb}(x,y)=(A-1)f(x,y)+f_{hp}(x,y)$
- $H_{hb}(u,v)=(A-1)+H_{hp}(u,v)$



a b
c d

FIGURE 4.29 Same as Fig. 3.43, but using frequency domain filtering. (a) Input image. (b) Laplacian of (a). (c) Image obtained using Eq. (4.4-17) with $A = 2$. (d) Same as (c), but with $A = 2.7$. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

An image formation model

- We can view an image f(x,y) as a product of two components:

$$f(x,y) = i(x,y) \cdot r(x,y)$$
$$0 < i(x,y) < \infty$$
$$0 < r(x,y) < 1$$

- i(x,y): illumination. It is determined by the illumination source
- r(x,y): reflectance (or transmissivity). It is determined by the characteristics of imaged objects

Homomorphic Filtering

- In some images, the quality of the image has reduced because of non-uniform illumination
- Homomorphic filtering can be used to perform illumination correction

$$f(x,y) = i(x,y) \cdot r(x,y)$$

- The above equation cannot be used directly in order to operate separately on the frequency components of illumination and reflectance

Homomorphic Filtering

ln : $z(x,y)=\ln f(x,y)=\ln i(x,y)+\ln r(x,y)$

DFT : $Z(u,v)=F_i(u,v)+F_r(u,v)$

H(u,v) : $S(u,v)=H(u,v)Z(u,v)$

$(\mathrm{DFT})^{-1}$: $s(x,y)=i'(x,y)+r'(x,y)$

exp : $g(x,y)=e^{s(x,y)}=i_0(x,y)r_0(x,y)$

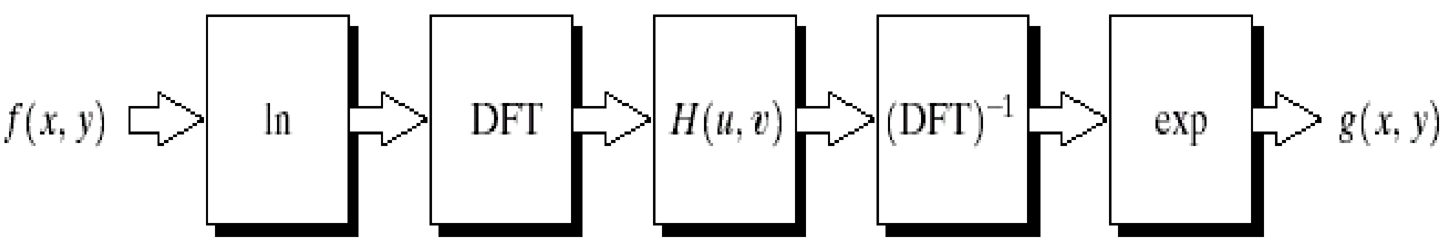


FIGURE 4.31 Homomorphic filtering approach for image enhancement.

Homomorphic Filtering

- By separating the illumination and reflectance components, homomorphic filter can then operate on them separately

- Illumination component of an image generally has slow variations, while the reflectance component vary abruptly

- By removing the low frequencies (highpass filtering) the effects of illumination can be removed

Homomorphic Filtering

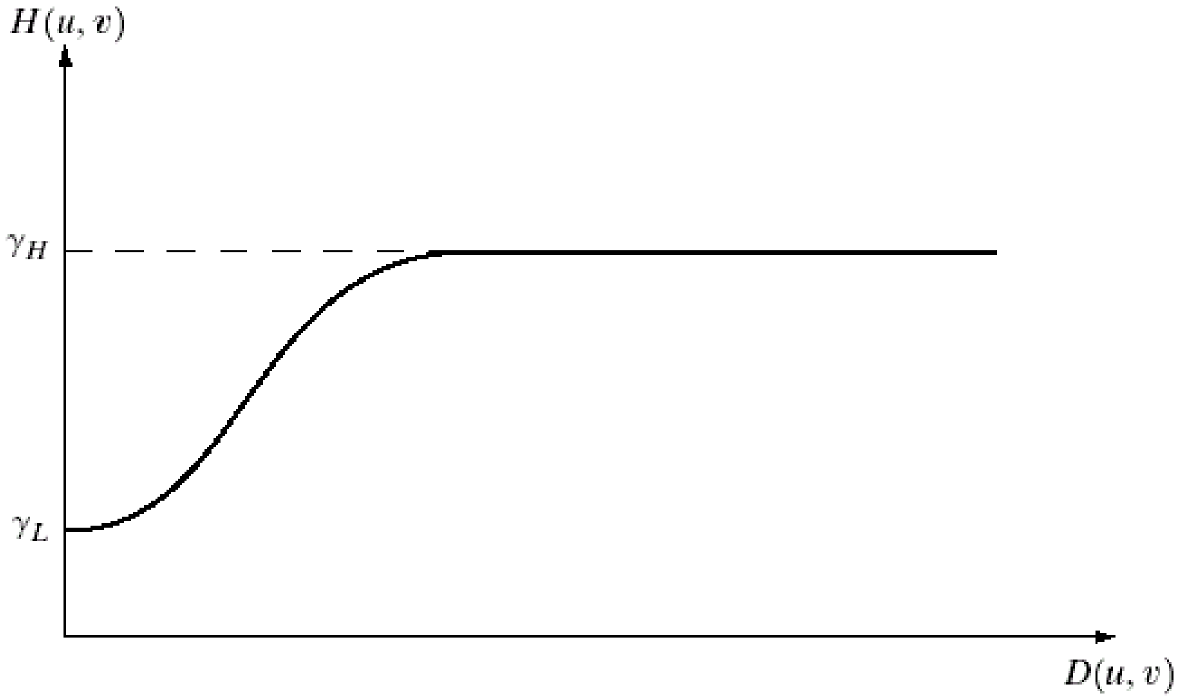
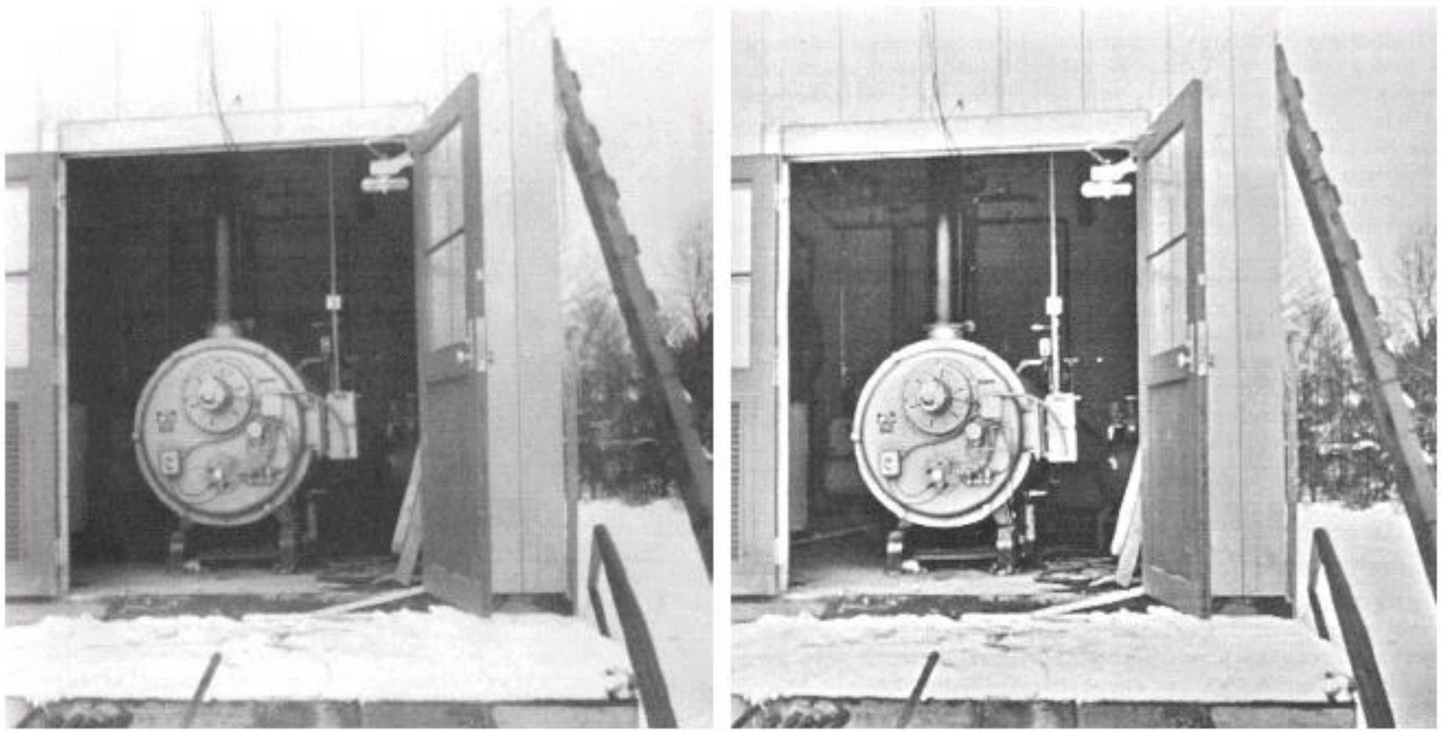


FIGURE 4.32 Cross section of a circularly symmetric filter function. $D(u, v)$ is the distance from the origin of the centered transform.

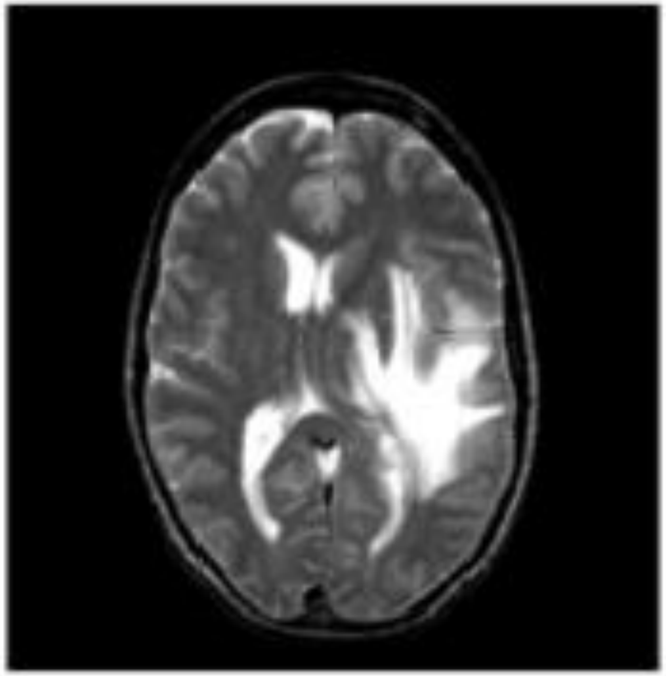
Homomorphic Filtering: Example 1



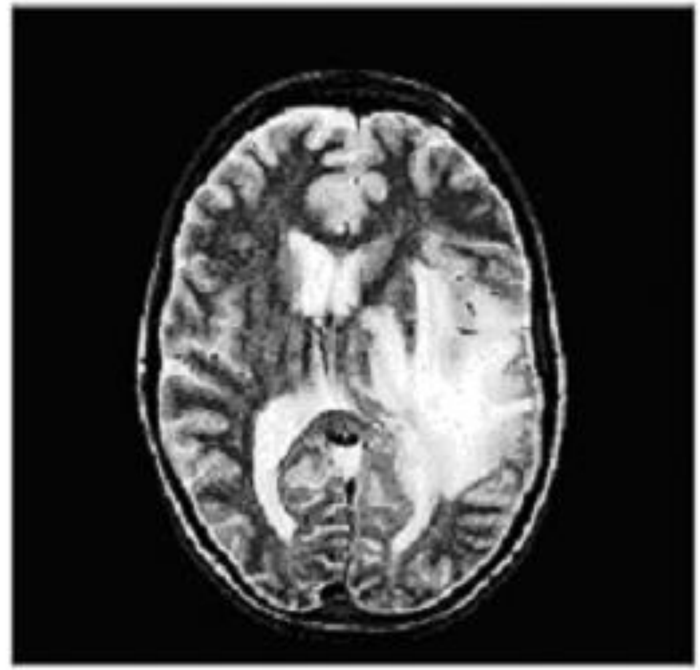
a b

FIGURE 4.33 (a) Original image. (b) Image processed by homomorphic filtering (note details inside shelter). (Stockham.)

Homomorphic Filtering: Example 2



Original image



Filtered image

THANK YOU